

AIMScribe — Software Requirements Specification

Integration specification for the CMED engineering team
AIMS LAB · Independent University, Bangladesh

Document	AIMS-SRS-001
Version	3.1
Date	14 September 2026
Status	Baseline for integration. Items marked OD-nn are open and need a decision (§14).
Plan described	The plan agreed on 13 September 2026 — AIMS LAB server at UIU, three CMED signals over two channels, every recording confirmed, a lossless copy kept in the cloud
Companion document	<i>AIMScribe — Integration Guide for CMED</i> (14 September 2026), the short version of what CMED builds
Recorder version	2.3.1 (<code>AIMScribe_Agent.exe</code>)
Wire protocol	2 today. The message changes in §6.1 are the next revision (§13.1).
Deployment	7 clinics · 14 consulting rooms · 30 doctors · 16 enrolled laptops

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Part I — Understanding the system

1. Introduction

1.1 What AIMSScribe is

AIMSScribe records the conversation between a doctor and a patient in a consulting room. AIMS LAB later transcribes the recording and uses it to draft prescriptions and to build a research dataset.

AIMSScribe can hear a consultation, but it cannot know whose consultation it is. Only CMED knows which doctor has opened which patient. So CMED tells AIMSScribe, with three short signals. Everything else — recording, encryption, upload, checking and storage — is built and run by AIMS LAB.

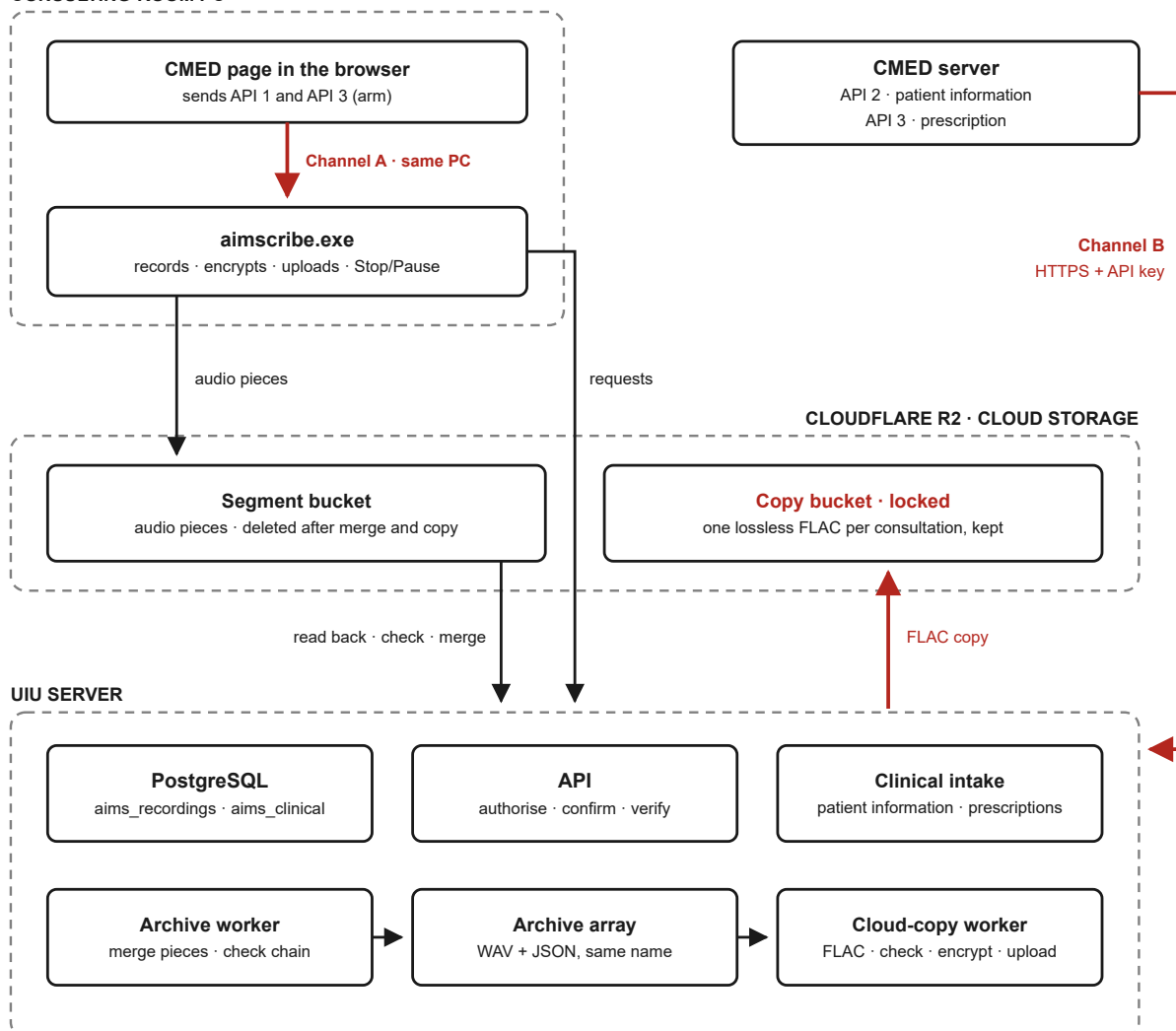
This document states what the whole system must do, as numbered requirements that can each be tested. It gives CMED a fixed contract to build against, and lets either side point to a requirement later and say whether it was met.

1.2 The system on one page

AIMScribe at a glance

Who sends what, and where the audio goes.

CONSULTING-ROOM PC



Audio travels only from the PC to Cloudflare R2 and the UIU server. It is never sent to CMED.

Each copy is deleted only after the next place has proved it holds a good copy (Figure 6).

Figure 1. AIMScribe at a glance. The CMED page tells the recorder on the same PC when a consultation starts and when its prescription is built (Channel A). CMED's server sends patient information and the prescription to the AIMS LAB server at UIU (Channel B). Audio pieces go from the PC to Cloudflare R2. The UIU server checks them, merges each consultation into one recording with its patient information beside it, and keeps a lossless copy in the cloud. Audio never goes to CMED.

One consultation, step by step.

#	What happens	Who does it
1	The doctor opens a patient's details in CMED.	Doctor
2	CMED's server creates <code>start_time</code> once and returns it to the page with the patient's details.	CMED server
3	The page sends API 1 — trigger to the recorder on the same PC. The microphone opens at once.	CMED page
4	At the same moment, CMED's server sends API 2 — patient information to the AIMS LAB server, with the same five fields and the patient's consent.	CMED server
5	The recorder asks the AIMS LAB server for permission to record. The server matches the request to API 2 and confirms the recording.	Recorder, AIMS LAB server
6	The recorder cuts the audio into 30–60 second pieces, encrypts each piece on disk, and uploads it to Cloudflare R2.	Recorder
7	The AIMS LAB server reads each piece back from R2 and checks its fingerprint. It then tells the PC it may delete its copy.	AIMS LAB server
8	The doctor builds the prescription. The page sends API 3 — prescription built to the recorder, and CMED's server sends the prescription to AIMS LAB. The recording keeps running while the doctor counsels the patient.	CMED page, CMED server
9	The doctor opens the next patient, and the page sends API 1 again. The previous recording closes and the next one opens at the same instant.	CMED page, recorder
10	The AIMS LAB server merges the pieces into one WAV file, checks it, and saves it on the UIU archive with a JSON file of the patient information beside it, under the same name.	AIMS LAB server
11	It compresses the WAV losslessly (FLAC), checks the copy, encrypts it and stores it in a locked cloud bucket. Only then are the pieces deleted from R2.	AIMS LAB server

Steps 1 to 4, 8 and 9 involve CMED. Steps 5 to 7, 10 and 11 are AIMS LAB's alone.

1.3 What CMED builds, and what AIMS LAB builds

CMED builds	AIMS LAB builds
A connection from the CMED page to the recorder on the same PC	The recorder, its installer, and the on-screen Stop and Pause control
API 1 — the trigger, when patient details open	Registering each PC, and authorising each recording
API 2 — patient information, server to server, at the same moment	Confirming each recording against API 2
API 3 — prescription built: the arm signal to the recorder, and the prescription to AIMS LAB	Upload, checking, the archive, the cloud copy and both databases
Handling replies quietly, so a recorder failure never affects the doctor	The UIU server, cloud storage, monitoring and support

CMED holds no cryptographic key, stores no audio, installs nothing on the doctor's PC, and builds no screen for AIMScribe. The only credential CMED holds is the API key for Channel B, kept on CMED's server. Estimated effort for CMED: **3–4 developer-days** (§13.2).

1.4 How to read this document

If you are	Read
A CMED developer building the integration	§1, §3, §6.1, §6.2, Appendices A, B and C
A CMED architect or tech lead	§1–§3, §5, §8.1, §11, §13, §14
A CMED security reviewer	§3.6, §4, §5, §6.2.1, §8.1, §8.4, §9.4
AIMS LAB engineering	All of it
UIU IT, or anyone buying hardware or storage	§10, §11, §14

Most of §7 to §11 describes AIMS LAB's own obligations. It is written down so CMED can see the whole system, and hold us to it.

1.5 Words used in this document

Term	Meaning
Recorder	AIMScribe_Agent.exe (called <code>aimscribe.exe</code> in the guides), the program on the doctor's PC that records. Also called <i>the agent</i> .
AIMS LAB server	The AIMS LAB backend, hosted on a server at UIU. Also called <i>the backend</i> or <i>the UIU server</i> .
Channel A	The connection from the CMED page to the recorder on the same PC, <code>ws://127.0.0.1:5050/ws</code> . It never leaves the machine.
Channel B	The connection from CMED's server to the AIMS LAB server, over HTTPS, with an API key.
API 1 • Trigger	Channel A message: a consultation has started. It opens the microphone.
API 2 • Patient information	Channel B message, sent at the same moment as API 1: the patient's consent, demographics, paramedic measurements, and the previous prescription.
API 3 • Prescription built	Two messages when the prescription is built: <code>prescription_built</code> to the recorder, which arms it, and the prescription contents to the AIMS LAB server.
The five fields	<code>patient_id</code> , <code>doctor_id</code> , <code>hospital_id</code> , <code>start_time</code> , <code>date</code> . API 1, API 2 and the prescription all carry them, identical.
Session	One recorded consultation, from the moment the microphone opens to the moment it closes.
Segment or piece	A 30–60 second part of a recording. Audio is sent in pieces, so a failure loses a piece rather than the whole recording.
Buffer	The recorder's encrypted store on the PC (also called the <i>spool</i>). Pieces wait here only until the server confirms a good copy.
Gate and arm	The rule that a new patient cannot end the current recording until its prescription has been built. API 3 arms the gate.
Grant	A one-minute, one-use permission to record one consultation, issued by the AIMS LAB server.
Confirmation	Matching a recording's permission request to the API 2 that CMED's server sent.
Enrolment	Registering one PC with AIMS LAB, once, before it is used.
Hash chain	A running, signed fingerprint of a recording. If any part is changed, removed or reordered, it stops matching.
Receipt	Proof from the server that it holds a verified copy of a piece. Only a receipt allows the PC to delete its copy.
Quarantine	Where a piece that fails its check is kept, outside the evidence archive, while it is reviewed automatically.
Cloudflare R2	The cloud storage service. It holds audio pieces until they are merged, and one lossless copy of each finished recording.
UIU archive	The disk array on the UIU server that holds each finished recording as a WAV file and a JSON file with the same name.
FLAC	A lossless audio format. It makes our recordings much smaller and decodes back to exactly the same samples.

Term	Meaning
Loopback	A connection between two programs on the same computer, <code>127.0.0.1</code> .
DPAPI	Windows Data Protection, used to protect secrets stored on the PC.
ULID	A unique, time-sortable identifier, used to name sessions.
PLP	Power-loss protection, a property of server-grade SSDs.
Idempotent	Doing the same thing twice has the same result as doing it once.

1.6 How requirements are written

Every requirement has an identifier such as `SRS-CAP-07`, a priority, a verification method and an owner. Identifiers are permanent. A withdrawn requirement is marked withdrawn and its number is never reused, so numbers are not always continuous.

Shall is binding. **Should** is a strong recommendation; leaving it out needs a written reason. **May** is optional.

Priority	Meaning
M	Mandatory. The system is not acceptable without it.
S	Should. Expected at general release; may be deferred through the pilot by agreement.
C	Could. Done only if it costs little.

Verification	Meaning
T	Automated test
D	Demonstration on a running system
I	Inspection of code, configuration or output
A	Analysis, calculation or measurement

Owner names who builds it: **AIMS**, **CMED**, or **Joint** where both sides change together. A requirement shown in a shaded block reads `SRS-GRP-nn` [priority, verification, owner], followed by its text.

1.7 Scope

In scope. Everything from the microphone to the archive and the cloud copy: recording on the PC; encryption and upload; the hash chain; checking each piece; starting, arming and closing a consultation; the on-screen control; registering PCs and authorising recordings; confirming recordings with CMED's server; receiving patient information and prescriptions; the two databases; the UIU server; cloud storage; and the hardware for all of it.

Out of scope. Speech recognition (ASR) and clinical entity extraction (NER). They run inside AIMS LAB after a recording is archived, and need nothing from CMED. Also out of scope: CMED's own application, database, screens and login. This document states what AIMScribe needs to *receive*, never how CMED should build what sends it.

1.8 What is new in version 3.0

This version brings the whole document in line with the plan agreed on 13 September 2026. Versions 2.0 to 2.3 added parts of that plan, but several sections still described the earlier one.

Topic	Earlier versions	Version 3.0
AIMS LAB server	Hosted on Render, database on Neon	A server at UIU , sized in §10.3
What CMED sends	Two signals to the recorder	Three signals over two channels — API 1, API 2 and API 3
The trigger	Carried a consent flag and a clinic code	Five fields: <code>patient_id</code> , <code>doctor_id</code> , <code>hospital_id</code> , <code>start_time</code> , <code>date</code>
Consent	Carried in the trigger	Carried in API 2 , from CMED's reception records. No consent, nothing kept.
The arm signal	<code>consultation_complete</code>	<code>prescription_built</code> , carrying the <code>session_id</code>
Is a request genuine?	Checked against AIMS LAB's register	Also confirmed against CMED's API 2 (§5.6)
Audio on the PC	Deleted after archiving, then a 24-hour wait	Deleted at once when the server holds a verified copy
Audio pieces in the cloud	Deleted after archiving	Kept until the archive is verified and a lossless FLAC copy is stored , then deleted
Patient data	Planned	Two databases , and a JSON file beside each recording with the same name
Sizing	A cloud hosting estimate	UIU server hardware and cloud storage , with measured compression

The full history is in Appendix E.

2. Overall description

2.1 The one hard problem

AIMScribe is a recorder that produces evidence. It is not a clinical system, and it makes no clinical decision.

Its central problem fits in one sentence: **AIMScribe can hear a consultation but cannot know whose it is.** A microphone captures sound. It does not capture which patient is present, which doctor is speaking, or that a consultation has begun. Only the clinical application the doctor is already using knows those things.

That is why CMED is involved. CMED is not asked to record, store or protect any audio. CMED is asked to say *this doctor has opened this patient* (API 1 and API 2), and later *the prescription is built* (API 3).

2.2 The three identities

Almost every design decision follows from keeping these three apart. Mixing any two of them caused misfiled recordings in the first version.

Identity	Where it comes from	Changes how often	Can a web page set it?
Clinic (<code>hospital_id</code>)	The PC's enrolment record at AIMS LAB	Never, for a given PC	No. The page's value is only checked against it.
Doctor (<code>doctor_id</code>)	CMED, per consultation	Twice a day — morning and afternoon shifts share a laptop	Only as a claim, checked against AIMS LAB's register and CMED's API 2
Patient (<code>patient_id</code>)	CMED, per consultation	Every consultation	Only as a claim, checked against CMED's API 2

Why the clinic comes from the machine. The clinic code is the top-level folder of the archive. If a page could set it, a mistyped or malicious value would create a new folder, and the recording would disappear into it without anyone noticing. It is fixed when an administrator enrolls the PC.

Why the doctor cannot come from the machine. A consulting room runs two shifts. Tying the doctor to the PC filed every afternoon consultation under the morning doctor — in the file name, where a wrong value looks most believable. The doctor must come from CMED, which knows who is logged in.

How CMED's `hospital_id` relates to ours. CMED sends its own clinic identifier in `hospital_id`, for example `CMED-DHK-BANANI-01`. AIMS LAB maps it to an internal code, for example `HOSP003`, using a table agreed once for each clinic (`SRS-ENR-19`). The internal code is what appears in file names and folders.

SRS-INV-01 [M, I, AIMS] The clinic under which a recording is filed shall come only from the device's enrolment record. A `hospital_id` received on any interface shall be checked against it, and never used in its place.

SRS-INV-02 [M, I, AIMS] A device's clinic code shall not change for the life of its enrolment. A clinic's *display name* may change at any time; its code may not, because existing archive paths depend on it.

SRS-INV-03 [M, T, AIMS] `doctor_id` shall have no default and no fallback anywhere in the system. A trigger naming no doctor shall be refused rather than attributed to a guess.

SRS-INV-04 [M, T, AIMS] A patient's name shall never be written to the hash chain, a file name, an archive path, a cloud object key or a log line. Names reach AIMS LAB only through API 2, and are stored only in the clinical database and in the JSON file beside the recording (§8).

2.3 Who uses the system

Group	Number	Technical skill	What they touch
Doctor	~30	Low. Uses CMED all day.	The Stop and Pause buttons, and nothing else. Never a terminal.
Clinic administrator	~7	Low to moderate	Runs the installer once per PC and pastes one code
AIMS LAB operator	2–3	High	Issues enrolment codes, watches alerts and the dashboard, runs the UIU server
CMED developer	2–4	High	Builds API 1, 2 and 3; never operates the system
Researcher or auditor	Occasional	Moderate	Reads recordings and checks chains; never changes anything

SRS-USR-01 [M, I, AIMS] No routine task performed by a doctor or a clinic administrator shall require a command line, a script, a `.bat` file or a terminal window. Clinical machines receive a signed installer and a graphical interface only.

A `.bat` file on a clinical PC is a support call waiting to happen, and an audit finding waiting to be written.

2.4 Operating environment

Element	Specification
Consulting-room PC	Windows 10 21H2 or Windows 11, x64, at least 4 GB RAM and 10 GB free disk
Audio input	USB speakerphone or microphone. How it is connected is a requirement, not a detail (§7.1a).
Browser	Chrome or Edge, current version, on the same PC as the recorder
Clinic network	Broadband where available, mobile tethering elsewhere. Short interruptions are expected.
Recorder	Python 3.12, delivered as one signed Windows installer; nothing else is installed
CMED server	Any technology able to make outbound HTTPS requests
AIMS LAB server	At UIU. Ubuntu Server 24.04 LTS, Docker Compose, an HTTPS gateway (Caddy or Nginx), FastAPI on Python 3.12
Databases	PostgreSQL 16 on the UIU server: <code>aims_recordings</code> and <code>aims_clinical</code>
Cloud storage	Cloudflare R2, S3-compatible: a segment bucket and a locked copy bucket
Archive	RAID 6 disk array on the UIU server: <code>/archive/<hospital_id>/<doctor_id>/<YYYY-MM-DD>/</code>
Time zone	<code>Asia/Dhaka</code> (UTC+06). Every timestamp on the wire is RFC 3339 with an explicit offset, for example <code>2026-09-13T10:14:32+06:00</code> .

2.5 Constraints

ID	Constraint	What follows from it
CON-01	CMED will not change its architecture for AIMScribe	The integration is additive: a small client in the page, and two outbound calls from CMED's server
CON-02	AIMS LAB never calls CMED	Every exchange starts at CMED: page to recorder (Channel A), and CMED server to AIMS LAB server (Channel B). CMED exposes no endpoint to AIMS LAB.
CON-03	CMED will not hold, generate or rotate a cryptographic key	Grants are issued by the AIMS LAB server (§5). The only credential CMED holds is the Channel B API key, which AIMS LAB issues and rotates.
CON-04	Clinical PCs get an installer, never scripts	<code>SRS-USR-01</code>
CON-05	Both AIMScribe source repositories are public	No secret may appear in either. Enrolment code sheets and the CMED API key are credentials.
CON-06	Clinic internet is limited and unreliable	Recording must continue through short outages; pieces wait in the buffer
CON-07	The archive is evidence	Lossy compression is never used, on the archive or on its cloud copy
CON-08	Consultations run back to back, 20–30 seconds apart	The handover between patients must have no gap
CON-09	The browser page is untrusted input	Not because CMED is distrusted, but because any page on the PC can open a connection. Every claim is checked.

2.6 Assumptions

ID	Assumption	If it proves false
ASM-01	The doctor's browser runs on the same PC as the recorder	Channel A cannot work; a different design is needed
ASM-02	CMED can run JavaScript on the patient-details page	No integration is possible
ASM-03	CMED holds a stable <code>patient_id</code> , <code>doctor_id</code> and clinic identifier	Recordings cannot be filed correctly
ASM-04	A Build Prescription action exists and is used for every patient	The gate has no signal; §7.7 and §11 cover its absence
ASM-05	Patient consent is obtained at reception, before the consultation, and recorded in CMED	Without it in API 2, nothing is kept (SRS-CNF-11)
ASM-06	CMED authenticates its own doctors	AIMScribe never authenticates anyone against CMED
ASM-07	PC clocks are within ± 5 minutes of true time	Grant checks fail; see §11
ASM-08	CMED's server can send HTTPS requests to the AIMS LAB server at the moment a patient is opened	Recordings stay unconfirmed until API 2 arrives
ASM-09	CMED's server can create <code>start_time</code> once and give the same value to the page and to API 2	Confirmation cannot match (§5.6)

2.7 Who is responsible for what

Concern	AIMS LAB	CMED
Audio capture, cutting, encryption and upload	•	
All cryptographic keys	•	
Registering PCs	•	
Authorising and confirming recordings	•	
UIU server, databases, archive and cloud storage	•	
The on-screen Stop and Pause control, and every message a doctor sees	•	
Recorder installation and support	•	
Server and storage capacity, and their cost	•	
Issuing and rotating the Channel B API key	•	
API 1 — the trigger, from the page		•
API 2 — patient information, from the server		•
Recording consent at reception, and sending it in API 2		•
API 3 — prescription built, from the page and the server		•
Creating <code>start_time</code> once, on the server		•
Handling replies, and failing quietly		•
Keeping the API key on the server, never in the page		•
Giving us the page addresses and clinic identifiers		•
Keeping identifiers stable over time		•
The message formats	①	①

3. System architecture

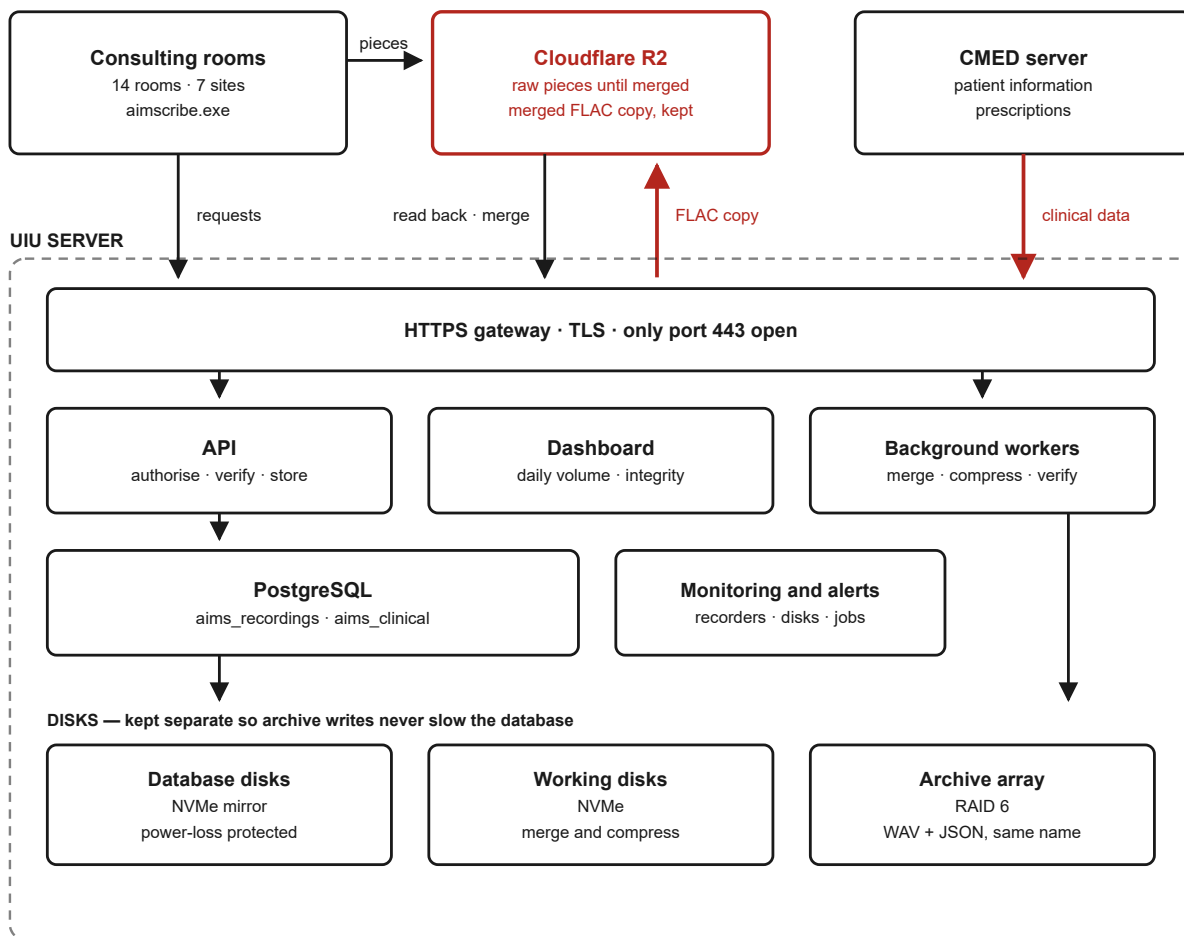
3.1 Components

#	Component	Runs on	Built with	Talks to
1	CMED page	Doctor's browser	CMED's stack plus a small client (Appendix B)	Recorder, over Channel A
2	CMED server	CMED	CMED's stack	AIMS LAB server, over Channel B
3	Recorder — control	Doctor's PC	FastAPI on <code>127.0.0.1:5050</code>	The page
4	Recorder — capture	Same program, own thread	PyAudio, WASAPI	The microphone
5	Recorder — cutter	Same program, own thread	NumPy	Capture to buffer
6	Recorder — buffer	Same program	AES-256-GCM and a journal	Local disk
7	Recorder — hash chain	Same program	Ed25519	Buffer, AIMS LAB server
8	Recorder — uploader	Same program	aiohttp	AIMS LAB server, Cloudflare R2
9	Recorder — Stop and Pause control	Same program, screen thread	tkinter	The doctor
10	HTTPS gateway	UIU server	Caddy or Nginx	The internet, on port 443 only
11	API	UIU server	FastAPI	Recorders, CMED server, workers
12	Clinical intake	UIU server, inside the API	FastAPI	CMED server, <code>aims_clinical</code>
13	Confirmation matcher	UIU server, inside the API	FastAPI	<code>aims_recordings</code>
14	Databases	UIU server	PostgreSQL 16 with PgBouncer	API, workers, dashboard
15	Segment bucket	Cloudflare R2	S3 API	Recorder (upload), API (read back), workers
16	Archive worker	UIU server	Python	Segment bucket, archive array, databases
17	Cloud-copy worker	UIU server	Python and FLAC	Archive array, copy bucket, databases
18	Copy bucket, locked	Cloudflare R2	S3 API	Cloud-copy worker
19	Dashboard	UIU server	Web application	<code>aims_recordings</code> only
20	Monitoring and alerts	UIU server	—	Everything above
21	<i>ASR and NER pipeline</i>	AIMS LAB	<i>out of scope</i>	Reads the archive

3.2 Where everything runs

AIMScribe on the UIU server

Pieces go to the cloud first. The merged, compressed copy stays there.



Nothing stays on clinic PCs, and no raw pieces stay in the cloud.

Each finished consultation: WAV + JSON on the archive array, and one lossless FLAC copy in R2.

Database dumps are also encrypted and sent to R2 every night.

Figure 2. AIMScribe on the UIU server. Recorders upload audio pieces to Cloudflare R2 and send their requests to the UIU server. The UIU server checks each piece, merges each consultation into one recording on its archive array, and uploads a lossless, encrypted copy back to R2. CMED's server sends clinical data to the same server.

Every connection is started by the side that has something to send. Recorders call the UIU server and R2. CMED's server calls the UIU server. The UIU server calls R2. Nothing ever calls into a clinic PC or into CMED. From the internet, the UIU server accepts connections on one port only: 443, for HTTPS.

Why the pieces go to R2 and not straight to UIU. Each recorder asks the UIU server for a short-lived upload link, then uploads the piece directly to R2. The UIU server reads the piece back to check it. This is how the system works today, and it is unchanged.

SRS-TOP-02 [M, I, AIMS] The UIU server shall accept connections from the internet on port 443 (HTTPS) only. Workers on the server shall make outbound connections only.

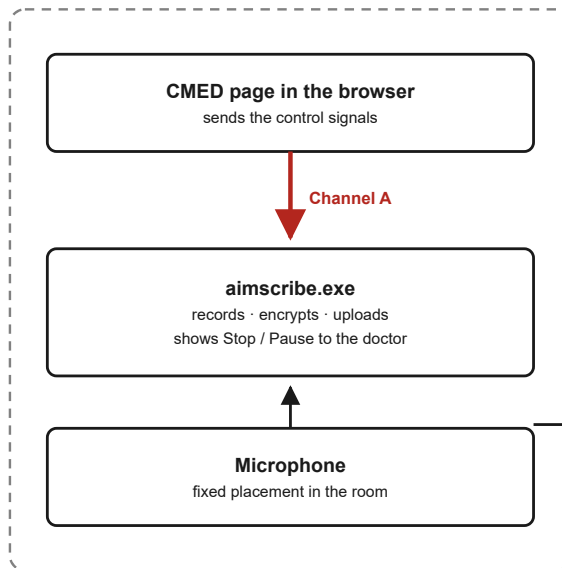
SRS-TOP-03 [M, I, Joint] AIMS LAB shall never open a connection to a service operated by CMED. Every exchange between the two organisations shall be started by CMED (CON-02).

3.3 Two channels

Two channels, two jobs

Control signals go to the PC. Clinical data goes server to server.

DOCTOR'S PC



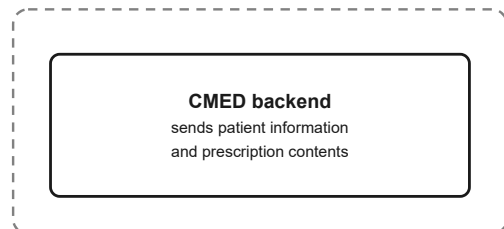
Channel A — to the PC

API 1 trigger, API 3 arm.
Small, instant, controls
the microphone.

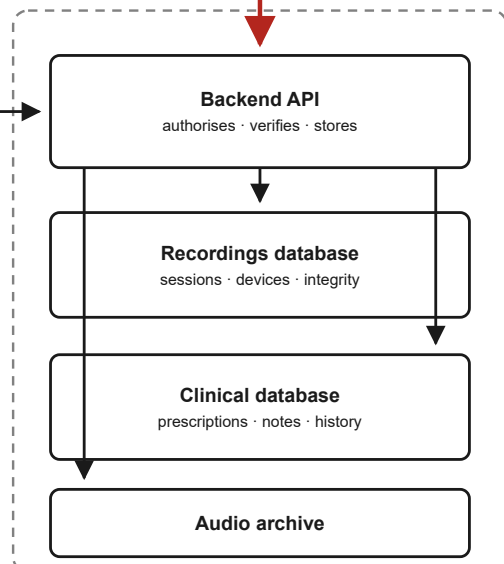
Channel B — to AIMS LAB

API 2 patient information,
API 3 prescription contents.
Server to server, no browser.

CMED SERVERS



AIMS LAB



Audio only ever travels from the PC to AIMS LAB.

It is never sent to CMED, in any form, at any time.

Figure 3. Two channels, two jobs. Control signals travel to the recorder on the doctor's own PC. Clinical data travels server to server, from CMED to AIMS LAB. Audio travels only from the PC to AIMS LAB, and never to CMED.

CMED sends two kinds of message, and they take different routes for a practical reason. Control signals must be instant, because they open a microphone. Clinical data can be large, and has no business passing through a browser.

	Route	Carries	Why this route
Channel A	CMED page → recorder, on the doctor's own PC	API 1 (trigger) and the arm half of API 3	It must be immediate. A round trip to a server before the microphone opened would lose the first seconds of the consultation — when the patient says why they came.
Channel B	CMED server → AIMS LAB server, over HTTPS	API 2 (patient information) and the prescription half of API 3	Server to server. Size does not matter, and nothing sensitive passes through a browser.

SRS-TOP-04 [M, I, Joint] Control signals shall travel on Channel A and clinical data on Channel B. Clinical data shall not be relayed through the browser, and control signals shall not depend on a server round trip.

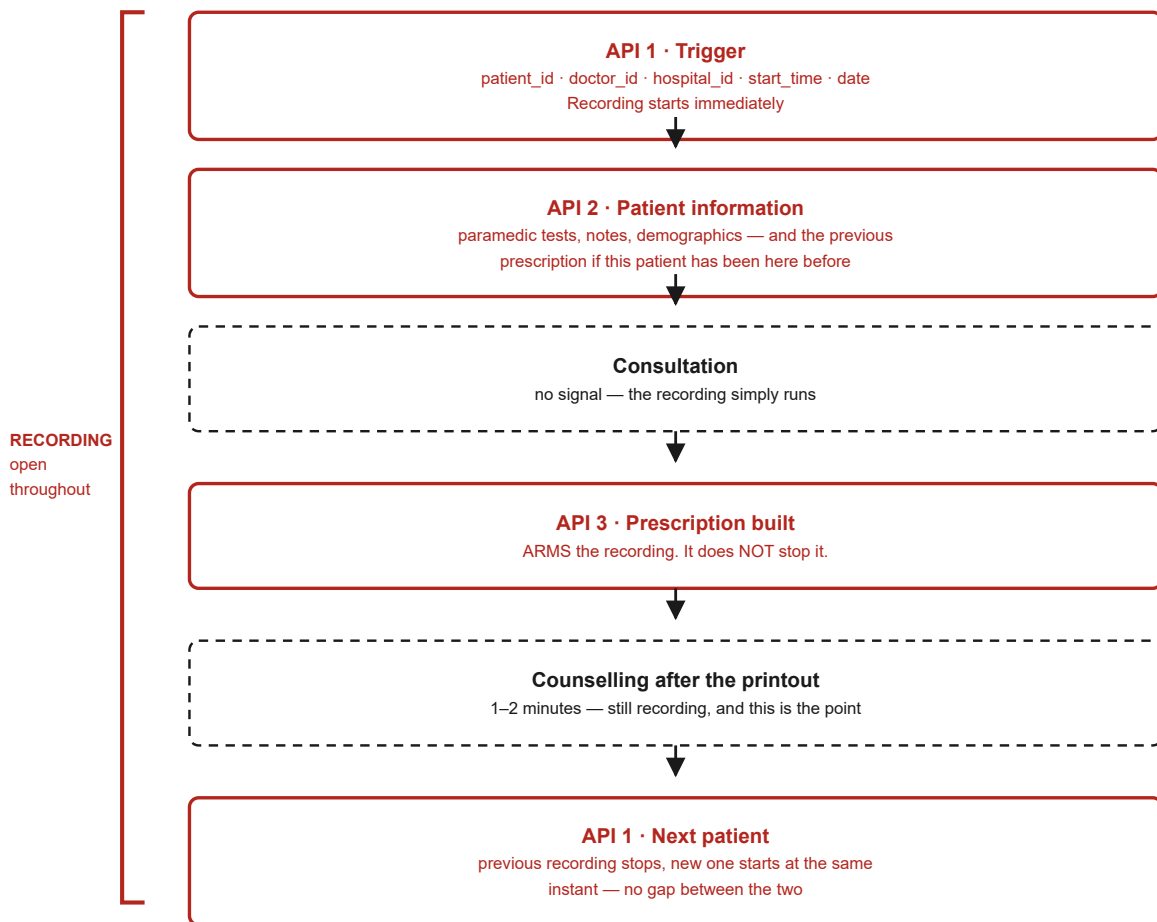
SRS-TOP-05 [M, I, AIMS] Channel B shall be authenticated with an API key issued to CMED, and shall accept clinical data only. No recording shall be started, stopped or paused over Channel B.

SRS-TOP-06 [M, T, AIMS] Audio shall never travel towards CMED, on either channel.

3.4 One consultation, start to finish

One consultation, three signals from CMED

The microphone is open continuously from the first trigger to the next patient's trigger.



Printing the prescription is not the end of the consultation.

That is why API 3 arms the recording instead of stopping it.

Figure 4. One consultation, three signals. The microphone stays open continuously from one patient's API 1 to the next patient's API 1. API 3 arms the recording; it does not stop it.

The point most easily missed is API 3. Printing the prescription is not the end of the consultation. The doctor prints it, hands it over, and then counsels the patient for another minute or two — and that counselling is exactly the part worth recording. So API 3 does not stop the recording. It *arms* it: this consultation is finished, so the next patient may now end it. The recording closes when the next patient's API 1 arrives.

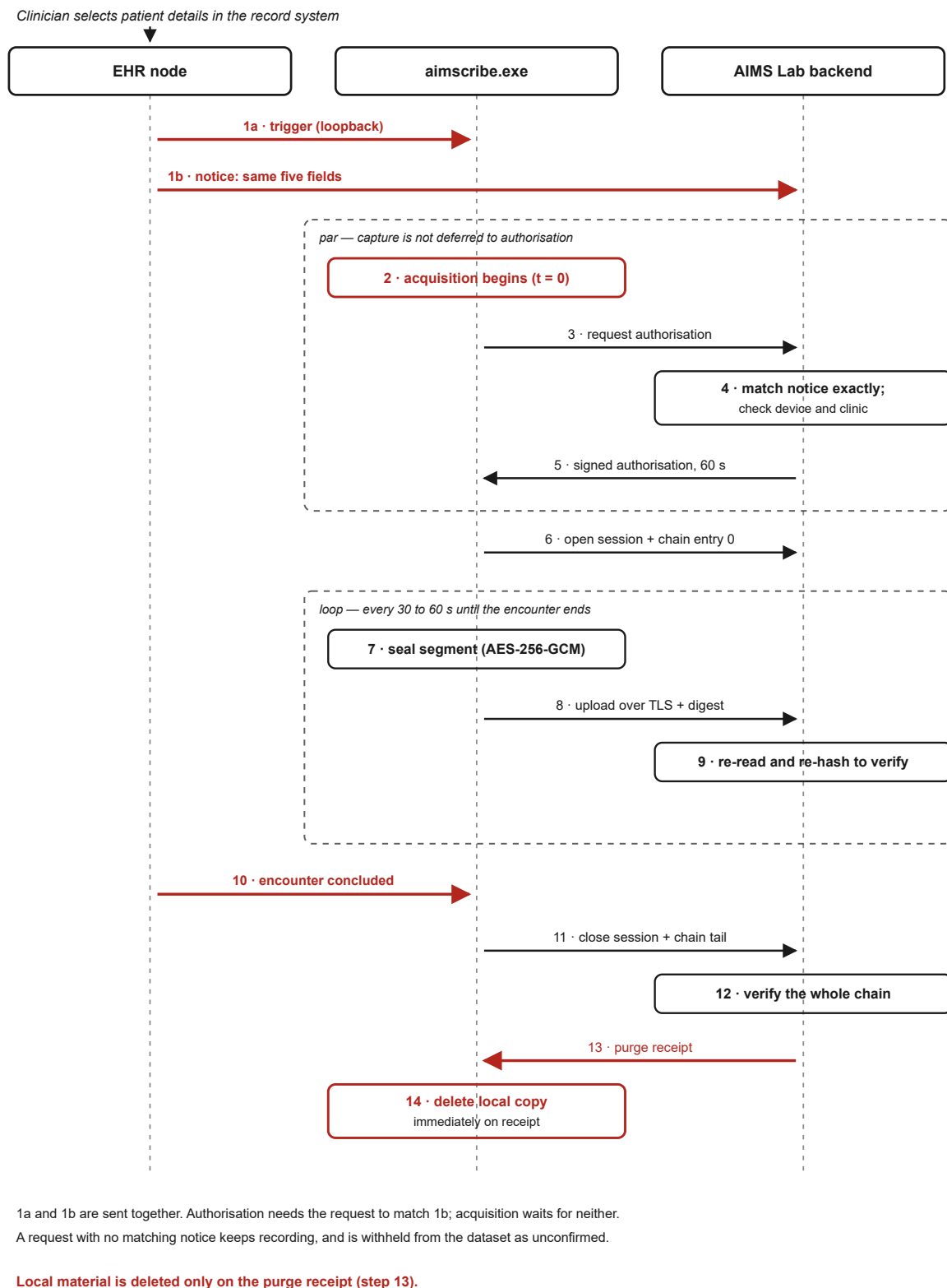
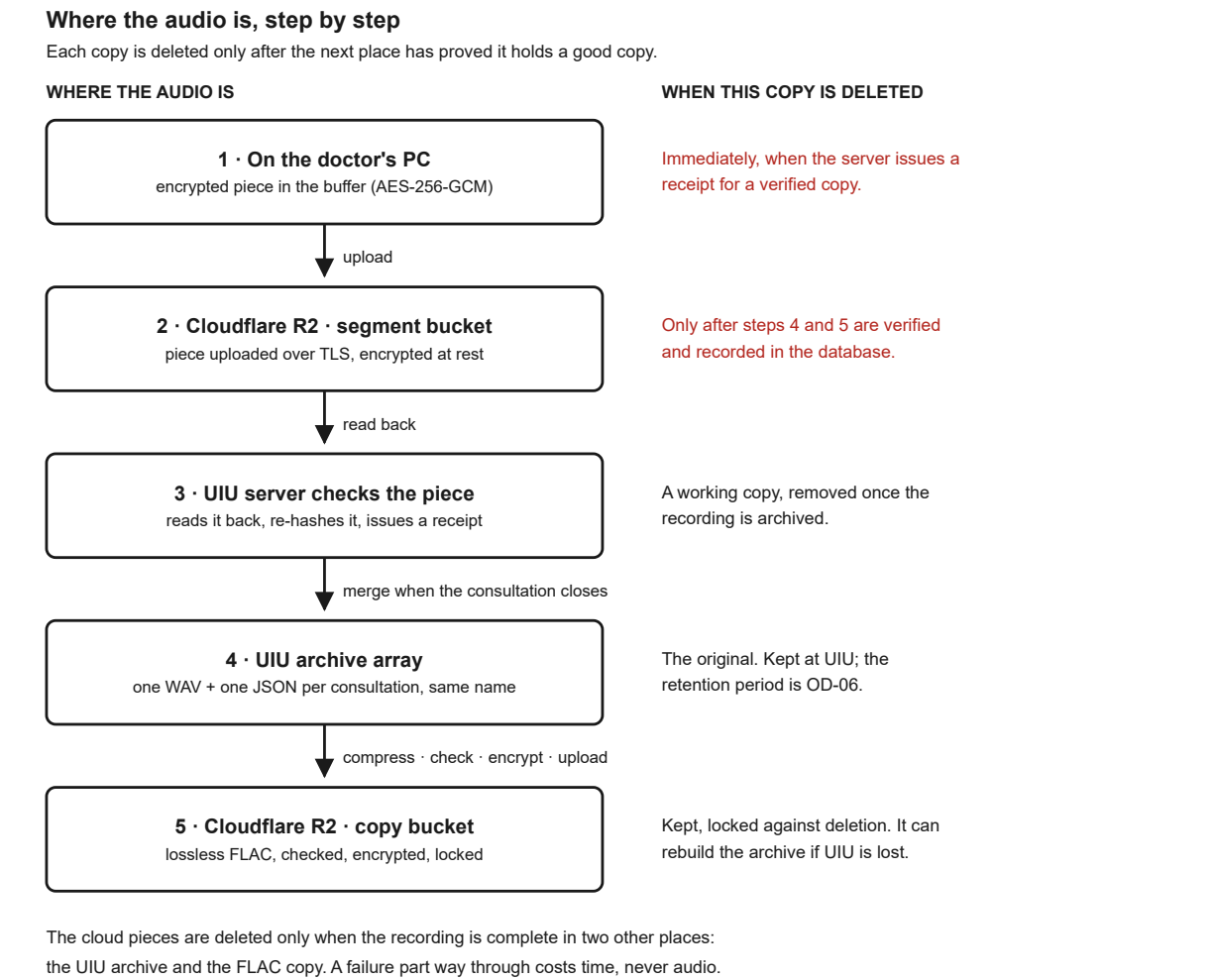


Figure 5. Behind the trigger. The trigger (1a) and the patient information (1b) leave CMED together. Recording starts at once, while the server matches the recorder's permission request to 1b. Pieces are sealed, uploaded and verified throughout, and the PC deletes its copy as soon as the receipt arrives.

Read steps 1a, 1b and 2 together. Recording starts before permission is confirmed, and the two run side by side. A slow network must never cost the opening seconds of a consultation. If permission is then refused for a hard reason — an unregistered PC, or the wrong clinic — the audio already captured is discarded (SRS-GRT-08).

The handover. When the next patient's trigger arrives and the gate is armed, the previous session closes and the next opens on separate threads, so there is no gap between two patients (CON-08).

3.5 Where the audio is, step by step



Why merging happens on the UIU server. R2 stores files; it cannot run the program that joins and compresses audio. The UIU server has already joined the pieces to make the archive WAV, so it compresses that same file and uploads it.

Why FLAC, and not a smaller lossy format. FLAC decodes back to exactly the samples in the archive, so the cloud copy can rebuild the archive and still be checked against the hash chain. A lossy copy such as Opus is about four times smaller again, but can do neither. Sizes and costs are in §10.4.

3.6 Trust boundaries

Party	Proves who it is with	Trusted to	Not trusted to
CMED page — really, any page on the PC	Nothing. Its web address is checked against an allowlist.	Say <i>where</i> a consultation is happening; on this PC	Choose the clinic, or start a trusted recording on its own
Recorder on an enrolled PC	A device token and a private key that never leaves the PC	Record, and sign what it recorded	Choose its own clinic, or record without a grant
CMED server	The Channel B API key	Say a doctor <i>really</i> opened a patient, say whether the patient consented, and send clinical data	Start, stop or control a recording
AIMS LAB server	Its TLS certificate and the grant signing key	Authorise recordings, verify audio, issue receipts	—

The trigger says where; the notice says whether. A page on the PC can reach the recorder. It cannot make CMED's server send a matching API 2. So a recording with no matching notice is never admitted to the dataset (§5.6).

SRS-TOP-01 [M, I, AIMS] The recorder shall listen on loopback addresses only. It shall never listen on an address reachable from the network.

Part II — The questions CMED asks first

4. Device enrolment and machine identity

Short answer: CMED builds nothing for enrolment, and maintains nothing. Enrolment is between the consulting-room PC and the AIMS LAB server. CMED is not part of it, cannot see it, and is not affected by it. This section explains why the integration cannot weaken it.

4.1 What enrolment is for

Enrolment exists for one purpose: **a laptop that has not been registered cannot record.** A PC not enrolled by an AIMS LAB administrator will install, start, show its tray icon — and refuse to record.

What enrolment does *not* do matters just as much, because it decides who builds what:

Enrolment does	Enrolment does not
Tie one PC to one clinic, permanently	Decide who the doctor is
Register the PC's public key with the AIMS LAB server	Authenticate a person
Give the PC a long-lived credential	Stop a random web page from asking the recorder to record
Give AIMS LAB a register of devices it can revoke	Involve CMED in any way

The third row leads to §5. Enrolment answers *may this machine record at all*. It does not answer *is this particular request to record genuine*. That is the job of the grant and the confirmation, and the two questions must not be confused.

4.2 The credentials

Five different secrets exist. Mixing them up is the most common cause of a misconfigured deployment, so here they all are.

#	Secret	Created by	Kept where	Lifetime	Use
1	Enrolment code (token)	AIMS LAB administrator, via <code>POST</code> <code>/api/v2/admin/enrollment-token</code>	An instruction sheet, then <code>%PROGRAMDATA%\AIMScribe\state\enrollment.token</code> until used, then deleted	72 hours by default, 720 at most, single use	Tur PC i enro dev.
2	Device token	AIMS LAB server, at enrolment	<code>%PROGRAMDATA%\AIMScribe\state\device.token</code> , protected by DPAPI	Until revoked	Eve req that
3	Device private key	The recorder, on first start. It never leaves the PC.	<code>%PROGRAMDATA%\AIMScribe\keys\</code> , protected by DPAPI	Until the machine is rebuilt	Sigr has enti
4	Grant signing key	AIMS LAB	UIU server configuration only	Rotatable	Sigr per recc
5	CMED API key	AIMS LAB administrator	CMED's server configuration only	Rotatable	Aut Cha (\$6.

SRS-ENR-01 [M, I, AIMS] The enrolment code shall be stored in the database as a SHA-256 hash only. The plain code shall exist nowhere but the instruction sheet, and cannot be recovered once that sheet is destroyed.

SRS-ENR-02 [M, T, AIMS] An enrolment code shall work once, enforced inside a single database transaction with row locking, so two simultaneous attempts cannot both succeed.

SRS-ENR-03 [M, T, AIMS] An enrolment code shall expire after between 1 and 720 hours, 72 by default.

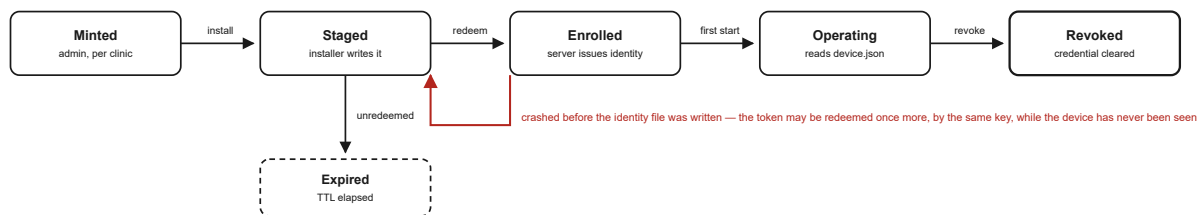
SRS-ENR-04 [M, I, AIMS] An unknown code, an expired code and an already-used code shall all produce the same error, so a caller learns nothing about which case occurred.

SRS-ENR-05 [M, I, AIMS] The device token and the device private key shall both be protected by DPAPI on disk. Unprotected storage shall be possible only under an explicit development setting that the production configuration check refuses.

SRS-ENR-06 [M, I, AIMS] Enrolment code sheets are credentials. They shall never be committed to a repository, and the generated `register.csv` shall contain no codes.

4.3 The life of an enrolment

One token per laptop, consumed once. The machine receives its clinic; it never asserts one.



A backend that is merely unreachable does not burn the token: it stays on disk and the agent retries at the next start. Burning it would strand a PC in another district with a credential that cannot be reissued remotely.

Figure 7. Device enrolment lifecycle. A code is created for a clinic, placed on the PC by the installer, used at first start, and replaced by a stored identity. An unreachable server does not use up the code, and a crash before the identity is saved returns the machine to the waiting state.

A server that is merely unreachable must not use up the administrator's code. If enrolment fails because the network is down, the code stays on disk and the recorder tries again at the next start. Using it up would strand a PC in another district with a credential that cannot be reissued remotely.

A code may be used twice, in one narrow case. Suppose the server records the enrolment, and the PC then fails to save its own copy — a full disk, a permissions problem, a crash between two writes. The PC now has no credential and its code is spent. The only symptom is a PC saying *not enrolled* while the server shows it enrolled. This happened in the field. So a spent code may be used once more, but only by the same machine, proven by the same device key, and only while that device has never been seen. After its first heartbeat, a second attempt is a replay and is refused.

SRS-ENR-07 [M, T, AIMS] An enrolment that fails because the server is unreachable shall leave the code on disk for a later retry.

SRS-ENR-08 [M, T, AIMS] A used code may be redeemed a second time only if all three hold: the device it created exists; that device has never been seen (`last_seen_at IS NULL`); and the public key presented matches the one registered. Any other repeat shall be refused and logged as a replay attempt.

SRS-ENR-09 [M, T, AIMS] The recorder shall save the device token before the identity file, so that a crash between the two leaves the machine unenrolled rather than holding an identity with no credential.

SRS-ENR-10 [M, T, AIMS] The recorder shall delete the enrolment code from disk immediately after it has been used successfully.

SRS-ENR-11 [M, T, AIMS] At every start, the recorder shall compare the stored key fingerprint with its live device key. If they differ — a wiped key folder, a restored disk image — it shall refuse to record and require re-enrolment.

4.4 How an administrator enrolls a PC

One code per **laptop**, not per doctor. A room with two laptops needs two codes.

1. The operator writes a **new, dated** list of the machines to enrol. Existing fleet files are never edited, because a rewritten `laptops.csv` reads as *these hospitals were deleted*.

2. A tool calls `POST /api/v2/admin/enrollment-token` once per machine and writes one instruction sheet per PC, plus a `register.csv` with no codes in it.
3. The clinic administrator runs `AIMScribeSetup.exe` on the PC and pastes the code into one of three fields. The other two are already filled in.
4. The installer places the code, sets the recorder to start at logon, and installs the pinned public keys. The recorder uses the code on first start and deletes it.
5. The instruction sheets are destroyed once the machines are working.

`doctor_id` on an enrolment record is **optional and decides nothing**. It labels the room in the paperwork. Most laptops are shared across shifts and have none. Even when one is set, it is never used to file a consultation, because the doctor arrives from CMED with every trigger (`SRS-INV-03`).

SRS-ENR-12 [M, I, AIMS] Fleet registration files shall only ever be added. A new dated file shall be written; an existing one shall not be rewritten.

SRS-ENR-13 [S, D, AIMS] Operators shall be able to create enrolment codes through an interface that does not require editing the database directly.

SRS-ENR-14 [M, I, AIMS] `doctor_id` on an enrolment code shall be optional, and shall never be used to file a recorded consultation.

4.5 Revoking a PC

SRS-ENR-15 [M, T, AIMS] An administrator shall be able to revoke a device by its identifier. Revocation shall delete the stored token hash, not merely set a flag, so a stolen laptop's credential stops working immediately.

SRS-ENR-16 [M, T, AIMS] A revoked device shall be refused at every authenticated route, and told clearly enough that its tray icon shows the state.

SRS-ENR-17 [M, I, AIMS] Every enrolment, re-issue and revocation shall be written to the append-only audit log with the acting administrator's name.

SRS-ENR-21 [M, T, AIMS] A device shall not be re-enrolled while pieces remain undelivered in its buffer. Re-enrolment creates a new device identity, and pieces from sessions opened under the old identity cannot be committed with the new credential. The buffer shall be emptied first, and the recorder shall refuse re-enrolment until it is.

4.6 What the integration changes for enrolment

Nothing. This is stated as a requirement rather than a reassurance, because CMED's security reviewer will ask.

SRS-ENR-18 [M, I, Joint] Enrolment shall require nothing from CMED. No CMED endpoint is called, no CMED credential is used, and no enrolment data is held by CMED. Enrolling, re-enrolling or revoking a PC shall need no change to CMED software.

Enrolment ties a **machine** to a **clinic**. Both are AIMS LAB facts: AIMS LAB owns the laptops, assigns each to a consulting room, and owns the archive whose top-level folder is the clinic code. A laptop moved to another clinic is an AIMS LAB logistics event, handled by re-enrolment, and CMED has nothing to do.

The **doctor** and the **patient**, by contrast, are CMED facts. They arrive with every consultation, in API 1 and API 2. None of CMED's messages is involved in registering a machine.

The one shared fact. For each clinic, AIMS LAB needs to know which CMED clinic identifier corresponds to which AIMS LAB clinic code. That is one row per clinic, agreed once in writing.

SRS-ENR-19 [M, I, Joint] For each clinic, CMED and AIMS LAB shall agree in writing one stable mapping from CMED's clinic identifier (sent as `hospital_id`) to the AIMS LAB clinic code, before the first recording at that site.

SRS-ENR-20 [M, T, AIMS] The AIMS LAB server shall refuse a trigger whose `hospital_id` does not map to the clinic of the device that sent it. §5.5 explains why it refuses rather than warns.

The clinic register. Seven sites, two consulting rooms each, one laptop per room:

AIMS LAB code	Site	Operator	Rooms	Status
HOSP001	Karail	Aalo	2	In service — signed recordings exist
HOSP002	Mirpur	Aalo	2	Assigned
HOSP003	Dholpur	Aalo	2	In service — signed recordings exist
HOSP004	Shyampur	Aalo	2	In service — signed recordings exist
HOSP005	Naryanganj	Aalo	2	To be assigned
HOSP006	Ershadnagar	Aalo	2	To be assigned
HOSP007	Amader Susastho	Amader Susastho	2	To be assigned

Fourteen rooms, fourteen laptops, and two spares held centrally: sixteen enrolment codes in total.

HOSP001, HOSP003 and HOSP004 can never be renamed. They appear inside signed hash-chain entries in the existing archive, so changing one would break the signature on every entry that follows it, and a signed entry cannot be edited by hand. Display names can still change at any time.

5. Authorisation, and confirming every recording

5.1 The gap enrolment leaves open

An enrolled laptop may record. That is not the same as *this request to record is genuine*.

Consider the gap concretely. A doctor's PC is enrolled and working. The doctor opens an unrelated website in another tab. That page runs JavaScript, opens a connection to `127.0.0.1:5050` — which any page on the machine can attempt — and sends a trigger. Without further checks, the recorder would record.

Two controls close that gap. The **grant** proves the AIMS LAB server has checked the request against its own register. The **confirmation** proves CMED's server really opened that patient.

5.2 What a grant is

A grant is a short-lived, single-use, signed statement from the AIMS LAB server that a named doctor may record a named patient at a named clinic.

Property	Value	Why
Algorithm	EdDSA (Ed25519)	Small, fast, and no parameters to get wrong
Lifetime	60 seconds	Long enough to reach the recorder; too short to be worth stealing
Clock allowance	5 seconds	PC clocks drift
Replay protection	A single-use identifier (<code>jti</code>), tracked by the recorder	A copied grant is refused even within its 60 seconds
Audience	<code>aimscribe-recorder</code>	A grant cannot be reused for another service
Claims	<code>patient_ref</code> , <code>doctor_id</code> , <code>hospital_id</code> , <code>start_time</code> , <code>consent_obtained</code>	The recording is tied to exactly one consented consultation

5.3 Where grants are made

In the dummy application AIMS LAB built for testing, the web page's own server made the grant and held a private key.

Real CMED will not do that, and does not need to. The grant is made by the AIMS LAB server. This is stronger than the dummy design, not weaker.

	Dummy application	Real CMED
Who holds a private key	The web application	The AIMS LAB server only
What the page sends	A signed token it made itself	Five plain fields
Who checks the doctor	Nobody	The AIMS LAB server, against its register
Who checks the clinic	Nobody	The AIMS LAB server, against the PC's enrolment
Who proves the consultation is real	Nobody	CMED's server, through API 2 (§5.6)
CMED's key-management work	Real	None

SRS-GRT-01 [M, I, AIMS] Grants shall be made only by the AIMS LAB server. No grant signing key shall be issued to, held by, or required of CMED.

SRS-GRT-02 [M, T, AIMS] The grant endpoint shall be authenticated by the calling device's token, and shall grant only for that device's enrolled clinic.

SRS-GRT-03 [M, T, AIMS] Before granting, the AIMS LAB server shall check that: the doctor exists and is active; the doctor works at that clinic; and the clinic matches the device's enrolment. Any failure shall refuse the grant with a distinct code. Whether a matching API 2 exists, and whether it records consent, is decided by §5.6.

SRS-GRT-04 [M, T, AIMS] The recorder shall check every grant against a public key pinned at installation: signature, issuer, audience and expiry, allowing no more than 5 seconds of clock difference.

SRS-GRT-05 [M, T, AIMS] The recorder shall accept each grant once, by `jti`, and refuse a repeat within its lifetime.

SRS-GRT-06 [M, T, AIMS] The recorder shall refuse to record if no grant verification key is installed. It shall never fall back to recording without authorisation under any setting reachable in production.

SRS-GRT-07 [M, D, AIMS] Asking for a grant shall not delay the microphone. Capture shall begin when the trigger arrives, and authorisation shall run alongside it.

SRS-GRT-08 [M, T, AIMS] If authorisation fails for a hard reason after capture has begun — the device is not enrolled or is revoked, the clinic does not match, the doctor is not registered at that clinic, or the matching API 2 records no consent — the audio captured under that trigger shall be discarded and not written to the buffer. A missing API 2 is not a hard failure; it is handled by §5.6.

5.4 What CMED builds for authorisation

Nothing cryptographic. CMED's part is:

1. Send five plain fields in API 1, instead of a signed token.
2. Create `start_time` once on the server, and use the same value in API 1 and API 2.
3. Send API 2 from the server at the moment the patient is opened, including the consent recorded at reception.
4. Read the recorder's reply and branch on `code`.

There is no key to generate, protect or rotate, no signing library, and no clock to synchronise. The only credential is the Channel B API key, which stays on CMED's server.

SRS-GRT-09 [M, I, CMED] CMED shall send the trigger as plain JSON fields, and shall not be required to compute any signature, hash or MAC.

5.5 A clinic mismatch is refused, not warned about

If a trigger names a clinic that does not match the PC's enrolment, one of two things is true: the clinic mapping is wrong, or a laptop is in a different building from where it is registered. Both produce wrongly labelled evidence, which is worse than no evidence, because it is believed.

SRS-GRT-10 [M, T, AIMS] A clinic mismatch shall refuse the recording with 401 CLINIC_MISMATCH and raise an operational alert. The system shall not record and warn.

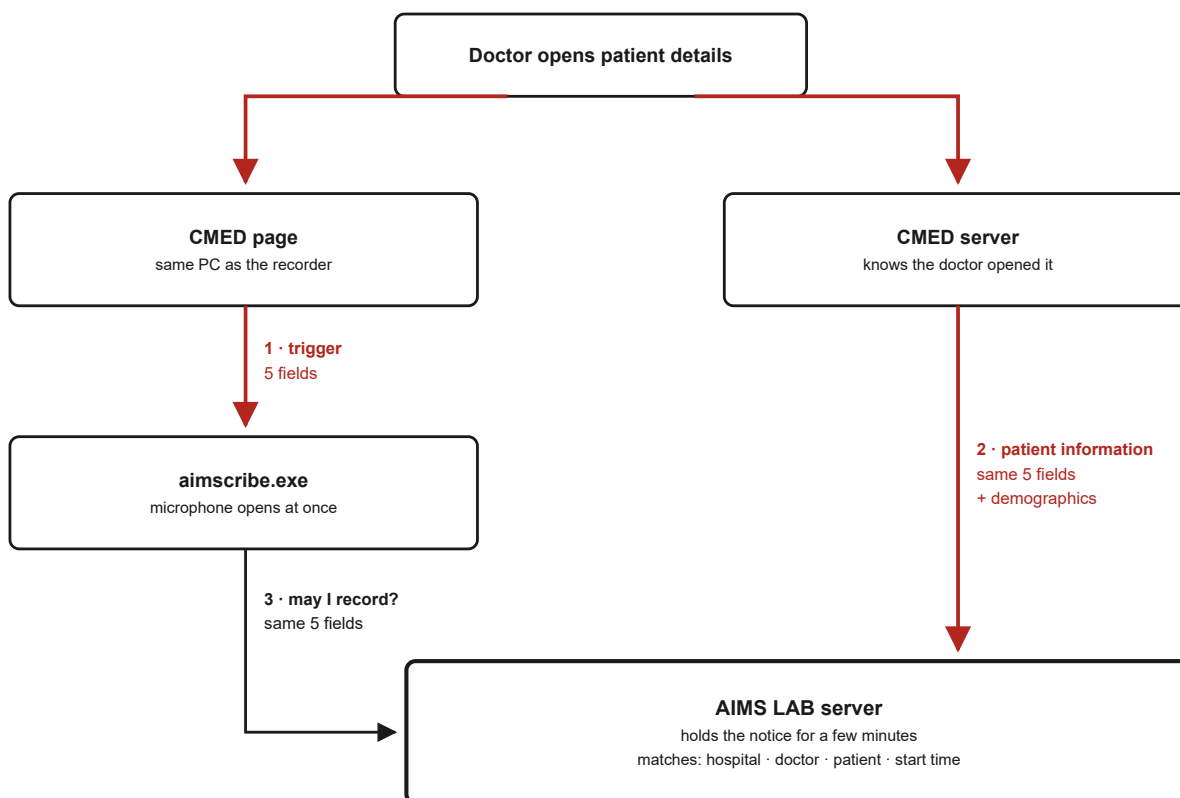
OD-03 offers CMED the option of recording with a warning during the pilot only.

5.6 Confirming every recording with CMED's server — CNF

The idea. When a doctor opens a patient, two messages leave CMED at the same moment, by two routes. The page sends API 1 to the recorder on the same PC (Channel A). CMED's server sends API 2 to the AIMS LAB server (Channel B). Both carry the same five fields. The AIMS LAB server confirms a recording only when the two match.

Two messages, two routes, one match

A recording is confirmed only when both copies of the same five fields meet at our server.



Match found — recording confirmed, and linked to its patient information.

No match — recording continues, marked unconfirmed, kept out of the dataset.

Figure 8. Two messages, two routes, one match. The page's trigger tells the recorder to start. CMED's server tells AIMS LAB the request is genuine. The recorder's permission request is matched to the notice on hospital, doctor, patient, date and start time.

Why it is needed. The page-address allowlist (§6.1.2) stops a page served from the wrong address. It cannot stop a page that has found a way past it. A matching API 2 can only come from CMED's server, which authenticates its own doctors. A page on the laptop can reach the recorder, but it cannot make CMED's server send AIMS LAB a notice.

Why the trigger still goes to the PC. The AIMS LAB server does not know which laptop a doctor is using, because rooms are shared across shifts. The page is running on that laptop, so the trigger lands on the right machine without any lookup. The trigger says *where*; the notice says *whether*.

Why the match can be exact. CMED's server creates `start_time` once, and the same value goes into both messages. Clock differences between the laptop and the servers therefore cannot affect the match. No `session_id` is needed in API 2.

Situation	What happens
API 2 arrives within a second or two	Confirmed. Nobody notices anything.
API 2 says the patient did not consent, or has no consent field	Recording stops at once. Audio already captured is deleted without being uploaded. The on-screen control tells the doctor this consultation is not being recorded. No patient information or prescription is kept.
API 2 is late	Recording continues. The recorder asks again every 5 seconds and shows "confirming". Confirmed when API 2 arrives.
API 2 never arrives	Recording is not cut. After two minutes it is marked <i>unconfirmed</i> , uploaded as usual, kept out of the dataset, and an alert is raised.
API 2 arrives after that	The recording is admitted.
No API 2 within 24 hours	The unconfirmed recording is deleted from AIMS LAB's storage, and the deletion logged.
The doctor opens a patient just to look	API 2 arrives, but no recording asks for it. It expires after five minutes. No harm.
The doctor opens the same patient twice	The most recent unused API 2 is matched.
The device is unregistered, or the clinic does not match	Refused, and the audio discarded (SRS-GRT-08).

SRS-CNF-01 [M, T, CMED] CMED's server shall send API 2 at the moment the patient is opened, at the same time as the page sends API 1. It shall not wait for the recording to start.

SRS-CNF-02 [M, T, Joint] API 2 and API 1 shall carry identical `patient_id`, `doctor_id`, `hospital_id`, `start_time` and `date`. `start_time` shall be created once, by CMED's server, and reused — never created separately by the page.

SRS-CNF-03 [M, T, AIMS] The AIMS LAB server shall hold each API 2 for matching for five minutes. A notice not matched within that time shall expire, and expiry shall not be treated as an error. The patient information itself is kept as a clinical record (§8.6).

SRS-CNF-04 [M, T, AIMS] A permission request shall be confirmed only when an unused API 2 matches it on all five fields, and that notice's hospital matches the enrolled clinic of the requesting device. A notice shall confirm at most one request.

SRS-CNF-05 [M, T, AIMS] Where more than one unused notice matches, the most recent shall be used.

SRS-CNF-06 [M, T, AIMS] The microphone shall not wait for confirmation. Capture begins on the trigger, and confirmation runs alongside it (**SRS-GRT-07**).

SRS-CNF-07 [M, T, AIMS] If no notice matches, the recorder shall ask again every five seconds and show "confirming" on its on-screen control. It shall not show an error to the doctor.

SRS-CNF-08 [M, T, AIMS] If no notice has matched after two minutes, the recording shall continue and be marked **unconfirmed**. It shall be uploaded as usual, kept out of the dataset, and an alert raised. It shall never be cut because a notice is late.

SRS-CNF-09 [M, T, AIMS] A notice arriving after a recording was marked unconfirmed shall still be matched, and the recording admitted. An unconfirmed recording that no notice matches within 24 hours shall be deleted from AIMS LAB's storage, and the deletion recorded in the audit log.

SRS-CNF-10 [M, T, AIMS] The match shall link the recording to its clinical record. The prescription sent later shall attach to the same consultation by the same five fields.

SRS-CNF-11 [M, T, AIMS] A recording shall be confirmed only when the matching API 2 carries `consent_obtained: true`. If it carries `false`, or no consent field, permission shall be refused with `CONSENT_NOT_GIVEN`, capture shall stop, and audio already captured under that trigger shall be deleted and never uploaded (**SRS-GRT-08**).

SRS-CNF-12 [M, T, AIMS] When API 2 records no consent, the AIMS LAB server shall keep no clinical content from it, or from the prescription for that consultation. It shall keep only the five fields and the time, for 24 hours, so it can refuse the recording and the prescription, and then delete them. Both requests shall be answered `200 CONSENT_NOT_GIVEN`.

A useful side effect. Because the match links each recording to its patient information from the moment it starts, the nightly reconciliation in §8.6 becomes a safety net, not the main way records and recordings are joined.

Part III — Requirements

6. Interfaces

This section defines every connection in the system. **§6.1 and §6.2 are what CMED builds.** §6.3 to §6.5 are AIMS LAB's own, shown so CMED's reviewers can see the whole chain of custody.

Interface	From → to	Built by	Section
Channel A	CMED page → recorder, same PC	CMED sends; AIMS LAB receives	§6.1
Channel B	CMED server → AIMS LAB server	CMED sends; AIMS LAB receives	§6.2
Recorder API	Recorder → AIMS LAB server	AIMS LAB	§6.3
Storage	Recorder and AIMS LAB server → Cloudflare R2	AIMS LAB	§6.4
Workers	Archive and cloud-copy workers	AIMS LAB	§6.5

6.1 Channel A — CMED page to the recorder

6.1.1 Connection

Property	Value
Address	<code>ws://127.0.0.1:5050/ws</code>
Health check	<code>GET http://127.0.0.1:5050/health</code> — no authentication, no side effects
Format	UTF-8 JSON, one object per WebSocket frame
Largest frame	64 KB; larger frames are refused
Reconnecting	Wait 1 second, doubling each time up to 30 seconds, for as long as the page is open
Waiting for a reply	3 seconds, then treat it as no reply
Origin header	Required; must exactly match an allowlisted page address

Why `127.0.0.1` and not `localhost`. On some Windows machines `localhost` resolves to the IPv6 address first. `127.0.0.1` always means this computer, and it is the address the recorder's security check expects.

Why plain `ws://` is allowed from an HTTPS page. Browsers treat the loopback address as a trustworthy origin (W3C *Secure Contexts*), so they do not block `ws://127.0.0.1` as mixed content. The alternative — a TLS certificate for `127.0.0.1` on every PC — is worse in every way: certificates expire, and an expired certificate on a clinic PC stops recordings with no warning.

Why WebSocket rather than HTTP. The recorder must be able to *push* news to the page — a stopped recording, a lost microphone — and plain HTTP cannot push. A WebSocket connection also lets the page know the recorder is alive without polling. There is deliberately no second way to start a recording, because that is where the origin check lives, and a second path would be a weaker one.

SRS-IF1-01 [M, D, CMED] The CMED page shall connect to `ws://127.0.0.1:5050/ws` from the doctor's browser and keep the connection open while the doctor is logged in.

SRS-IF1-02 [M, T, CMED] The page shall reconnect after a disconnection with a growing delay, and shall never retry in a tight loop.

SRS-IF1-03 [M, I, CMED] If CMED uses a Content-Security-Policy, its `connect-src` directive shall allow `ws://127.0.0.1:5050`. Without it, the browser blocks the connection before the recorder sees it.

SRS-IF1-04 [S, D, CMED] The page should call `GET /health` before its first connection, to tell *recorder not installed* apart from *recorder failing*.

6.1.2 Who may connect

Three checks run in order **before** the connection is accepted. Each failure closes the socket with code `4403`.

#	Check	Rule	Protects against
1	Origin	Must appear exactly in the allowlist. A missing or <code>null</code> origin is refused.	A random page on the PC opening a connection
2	Host	Must be an expected loopback address	DNS rebinding — a hostile site name made to point at <code>127.0.0.1</code>
3	Caller address	Must be <code>127.0.0.1</code> or <code>::1</code>	A caller from the network

Check 3 alone was the flaw in the first version: every browser on the PC connects from the loopback address, so it always passed.

SRS-IF1-05 [M, I, CMED] CMED shall give AIMS LAB the exact page addresses — scheme, host and port — from which the page will connect, for production and for testing. Wildcards shall not be accepted.

SRS-IF1-06 [M, T, AIMS] The recorder shall refuse a connection whose origin is missing, `null`, or not exactly on the allowlist, and shall refuse to start with an allowlist containing a wildcard.

6.1.3 API 1 — the trigger

When: the doctor opens a patient's details or history.

```
{
  "command": "start",
  "request_id": "cmed-7f3a2b91",
  "trigger": {
    "patient_id": "P0012345",
    "doctor_id": "DR0042",
    "hospital_id": "CMED-DHK-BANANI-01",
    "start_time": "2026-09-13T10:14:32+06:00",
    "date": "2026-09-13"
  }
}
```

Field	Required	Rule
patient_id	Yes	Letters, digits, _ and - only, up to 64 characters. Always the same value for the same patient.
doctor_id	Yes	Same rule. Never blank — the recorder refuses rather than guesses (SRS-INV-03).
hospital_id	Yes	Same rule. CMED's own clinic identifier; must map to this PC's clinic (SRS-ENR-19).
start_time	Yes	RFC 3339 with the offset, e.g. +06:00 . Created by CMED's server , identical to API 2 (SRS-CNF-02).
date	Yes	YYYY-MM-DD , the clinic's local date
request_id	Helpful	Any string. It is sent back in the reply so the page can match it.

Why patient_id must never change for a patient. It becomes part of the recording's file name and links every recording to that patient's clinical record. If it changes, the patient's recordings split into two unrelated sets that cannot be joined again.

6.1.4 The reply

Every command receives exactly one reply, with a numeric status that follows HTTP meanings and a fixed text code .

```
{
  "event": "ack",
  "command": "start",
  "request_id": "cmed-7f3a2b91",
  "status": 200,
  "code": "RECORDING_STARTED",
  "message": "Recording started.",
  "data": {
    "session_id": "01JB8XQ4M7YZ2K9V3N5P6R8T0W",
    "started_at": "2026-09-13T10:14:32+06:00",
    "armed": false
  }
}
```

The page keeps session_id and sends it back with prescription_built (§6.1.5). CMED's server does not need it.

SRS-IF1-07 [M, T, AIMS] Every command shall receive exactly one reply carrying status , code and, where the page supplied one, its request_id .

SRS-IF1-08 [M, D, AIMS] A recording that has started shall reply status: 200 , code: "RECORDING_STARTED" , with the session_id .

SRS-IF1-09 [M, I, CMED] The page shall act on code , never on message . Message text is written for doctors and may be reworted at any time; codes do not change.

SRS-IF1-10 [M, T, AIMS] The full set of codes in Appendix A shall be implemented, and no other code shall be returned for these commands.

6.1.5 API 3, part one — `prescription_built`

When: the doctor presses **Build Prescription** and it succeeds. This happens for every patient, because the paramedic in the investigation room reads the result, so it is a reliable sign that the consultation has reached its end.

```
{
  "command": "prescription_built",
  "request_id": "cmed-7f3a2b92",
  "patient_id": "P0012345",
  "session_id": "01JB8XQ4M7YZ2K9V3N5P6R8T0W",
  "occurred_at": "2026-09-13T10:26:11+06:00"
}
```

What this signal does, and does not do. It does **not** stop the recording. The doctor builds the prescription, prints it, hands it over, and then counsels the patient for another minute or two — valuable audio. The signal *arms the gate*: it allows the **next** trigger to end this session. Recording continues until the next patient is opened, typically 20–30 seconds later.

At the same moment, CMED's server sends the prescription contents over Channel B (§6.2.3).

SRS-IF1-11 [M, D, CMED] The page shall send `prescription_built` when the Build Prescription action succeeds, carrying the same `patient_id` as the trigger and the `session_id` from the trigger's reply.

SRS-IF1-12 [M, T, AIMS] `prescription_built` shall not stop, pause or otherwise change the current recording. It shall only arm the gate.

SRS-IF1-13 [M, T, AIMS] A `prescription_built` naming a patient or session other than the open session's shall be refused with 409 PATIENT_MISMATCH and shall not arm the gate.

SRS-IF1-14 [M, T, AIMS] A repeated `prescription_built` for a session already armed shall succeed again, with 200 GATE_ALREADY_ARMED .

SRS-IF1-15 [M, I, AIMS] The recorder shall use `prescription_built` only to arm the gate. Nothing in it shall be stored as clinical content; the prescription itself arrives over Channel B.

6.1.6 Messages the recorder sends without being asked

Event	When	What CMED should do
<code>status</code>	On connection, and whenever state changes	Update an indicator, if CMED shows one
<code>session_started</code>	Recording is running	Nothing required
<code>session_stopped</code>	Session closed, with the reason — including <code>consent_not_given</code>	Nothing required
<code>session_paused</code> · <code>session_resumed</code>	The doctor used the on-screen control	Nothing required
<code>session_confirmed</code> · <code>session_unconfirmed</code>	Result of the confirmation (§5.6)	Nothing required
<code>gate_armed</code>	<code>prescription_built</code> accepted	Nothing required
<code>warning</code>	Disk low, microphone lost, server unreachable	Nothing required

SRS-IF1-16 [M, I, AIMS] **CMED shall not be required to display anything.** Every message meant for the doctor — recording state, gate state, "confirming", level prompts, disk and microphone warnings, errors — shall come from the AIMScribe on-screen control (§7.8), which AIMS LAB builds.

SRS-IF1-18 [C, D, CMED] CMED *may* show its own indicator from these events, but nothing in this specification depends on it, and the system works fully when CMED shows nothing.

This follows from CMED's constraint not to change its software beyond sending the signals. If doctor-facing messages depended on CMED's screens, every new message would need a CMED release. Because the on-screen control is ours, AIMS LAB can change what a doctor is told without asking anyone.

6.1.7 When the recorder is absent

This requirement outranks the rest of the document.

SRS-IF1-17 [M, T, CMED] Any failure in the AIMScribe integration — connection refused, timeout, error reply, malformed message, or a Channel B call that fails — shall be caught, logged and otherwise ignored by CMED. It shall never show a dialog, block navigation, delay the patient's details, or stop the doctor seeing the patient.

A recording that fails is a recording lost. A clinical system that stops because a recorder failed is a patient not seen. These are not comparable, and the design never treats them as such. In code it is one line: `.catch(() => {})` (Appendix B).

6.2 Channel B — CMED server to the AIMS LAB server — CHB

6.2.1 Connection

Property	Value
Base address	<code>https://<aims-lab-server>/api/v2/clinical/</code> — the address is given to CMED at the start (for example <code>https://aimscribe.uiu.ac.bd</code> , to be confirmed with UIU IT, OD-18)
Authentication	Header <code>X-CMED-Key: <key issued by AIMS LAB></code>
Transport	HTTPS, TLS 1.2 or later
Format	<code>Content-Type: application/json</code> , UTF-8
Largest request	1 MB
Endpoints	<code>POST patient-information</code> (API 2) · <code>POST prescription</code> (API 3, part two)
Test environment	A separate address and a separate key (<code>SRS-API-05</code>)

SRS-CHB-01 [M, I, CMED] Channel B requests shall be sent from CMED's server only. The API key shall never be sent to a browser, placed in a web page, logged, or committed to a repository.

SRS-CHB-02 [M, T, AIMS] The AIMS LAB server shall refuse a request with a missing or wrong key with `401 INVALID_KEY`, and store nothing from it.

SRS-CHB-03 [M, D, AIMS] AIMS LAB shall issue separate keys for production and testing, and shall be able to replace a key with an overlap period in which both old and new work, so a key can be changed without a coordinated release.

SRS-CHB-04 [M, T, AIMS] The AIMS LAB server shall reply as soon as the request is safely stored — `202 ACCEPTED` — and within one second at the 95th percentile. Loading the record into the database may continue after the reply.

SRS-CHB-05 [M, T, CMED] A request that fails with a timeout, a network error, `429` or a `5xx` status shall be retried in the background with a growing delay, for at least 24 hours. A `4xx` other than `429` shall not be retried unchanged; it shall be logged for a developer.

SRS-CHB-06 [M, T, CMED] Sending or retrying a Channel B request shall never delay showing the patient's details to the doctor, or building the prescription.

SRS-CHB-07 [M, T, AIMS] Both endpoints shall be idempotent. The same request sent twice shall produce one record, and the second reply shall be `200 ALREADY_RECEIVED` (**SRS-CRI-09**).

SRS-CHB-08 [M, T, AIMS] A prescription sent again for the same five fields with different contents — for example, after the doctor edits and rebuilds it — shall be stored as a new version. The latest version is current; earlier versions are kept.

SRS-CHB-09 [M, T, AIMS] A prescription that arrives before its patient information shall be accepted, and attached by the five fields when the patient information arrives.

SRS-CHB-10 [M, T, AIMS] A request that fails validation shall be stored unchanged in a quarantine area and answered `422 SCHEMA_INVALID`, naming the fields at fault (**SRS-CRI-07**).

SRS-CHB-11 [M, T, AIMS] Fields AIMS LAB does not yet use shall be stored, not rejected (**SRS-NFM-05**), so CMED can send more than is listed without breaking anything.

SRS-CHB-12 [S, I, Joint] The list of fields for each endpoint shall be agreed in writing and given a version number (**SRS-CRI-11**).

6.2.2 API 2 — patient information

When: the moment the doctor opens the patient's details — at the same time the page sends API 1. Do not wait for the recording to start (**SRS-CNF-01**).

Why it is needed. Before the patient enters the room, reception asks for the patient's consent and a paramedic records their details and takes basic measurements. The doctor needs these at the start of the consultation, and so does AIMS LAB. For a returning patient, AIMS LAB also needs the most recent prescription: the doctor refers to it, and the transcript cannot be understood without it. API 2 is also what confirms the recording is genuine (§5.6).

```
POST /api/v2/clinical/patient-information
X-CMED-Key: <the key AIMS LAB issues to CMED>
Content-Type: application/json
```

```
{
  "patient_id": "P0012345",
  "doctor_id": "DR0042",
  "hospital_id": "CMED-DHK-BANANI-01",
  "start_time": "2026-09-13T10:14:32+06:00",
  "date": "2026-09-13",
  "consent_obtained": true,
  "consent_method": "verbal_at_reception",
  "consent_recorded_at": "2026-09-13T10:02:00+06:00",
  "demographics": {
    "name": "...", "sex": "female", "age_years": 34,
    "phone": "...", "address": "..."
  },
  "paramedic": {
    "recorded_at": "2026-09-13T10:06:00+06:00",
    "weight_kg": 58.0, "height_cm": 156,
    "blood_pressure": "120/80", "pulse_bpm": 78,
    "temperature_c": 37.1, "spo2_percent": 98,
    "notes": "..."
  },
  "previous_visit": {
    "date": "2026-06-02",
    "prescription": { "...": "..." },
    "diagnoses": [ "..."],
    "notes": "..."
  }
}
```

Field	Required	Rule
The five fields	Yes	Identical to API 1, character for character
demographics	Yes	name , sex (female or male), age_years or date_of_birth , phone , address — send what CMED holds
paramedic	When measured	recorded_at and each measurement taken. The full list is agreed with CMED (OD-16).
previous_visit	Yes	null for a first visit. Otherwise the date, prescription, diagnoses and notes of the most recent visit to this hospital.
consent_obtained	Yes	true only if the patient agreed, at reception, to be recorded and to their record being used. false or missing: nothing is kept (SRS-CNF-11 , SRS-CNF-12).
consent_method	Yes	How consent was taken, up to 64 characters — for example verbal_at_reception or signed_form
consent_recorded_at	Helpful	When reception recorded the consent, RFC 3339

The fields needed for female and male patients differ. CMED sends what applies; the clinical database enforces the right rules for each (SRS-DBA-06). Send everything held for a returning patient — a field AIMS LAB does not yet use is stored, not lost.

Reply:

```
{ "status": 202, "code": "ACCEPTED", "record_id": "01JB8XR2K4...",  
  "received_at": "2026-09-13T10:14:33+06:00" }
```

6.2.3 API 3, part two — the prescription

When: the doctor presses Build Prescription and it succeeds — the same moment the page sends `prescription_built`.

```
POST /api/v2/clinical/prescription  
X-CMED-Key: <your key>  
Content-Type: application/json
```

```
{  
  "patient_id": "P0012345",  
  "doctor_id": "DR0042",  
  "hospital_id": "CMED-DHK-BANANI-01",  
  "start_time": "2026-09-13T10:14:32+06:00",  
  "date": "2026-09-13",  
  "issued_at": "2026-09-13T10:26:11+06:00",  
  "diagnoses": [ "..."],  
  "items": [  
    { "drug": "...", "dose": "...", "frequency": "...", "duration": "...",  
      "instructions": "..." }  
  ],  
  "investigations": [ "..."],  
  "advice": "...",  
  "follow_up": "2026-10-13",  
  "notes": "..."  
}
```

Field	Required	Rule
The five fields	Yes	The same as API 1, with the consultation's original <code>start_time</code> — not the time the prescription was issued
<code>issued_at</code>	Yes	When the prescription was built
<code>diagnoses</code>	Yes	A list; empty if none
<code>items</code>	Yes	A list, one entry per medicine , each with <code>drug</code> , <code>dose</code> , <code>frequency</code> , <code>duration</code> , <code>instructions</code>
<code>investigations</code>	Yes	A list of tests ordered; empty if none
<code>advice</code> , <code>follow_up</code> , <code>notes</code>	When present	<code>follow_up</code> is a date, YYYY-MM-DD

Why items must be a list. AIMS LAB stores each medicine as its own row, with dose and frequency in their own columns. A block of text has to be taken apart later, and information that was clear when CMED sent it is lost in the process.

6.2.4 Channel B replies

status	code	Meaning	What CMED's server should do
202	<code>ACCEPTED</code>	Stored	Nothing
200	<code>ALREADY_RECEIVED</code>	This exact request was already stored	Nothing
200	<code>CONSENT_NOT_GIVEN</code>	The patient did not consent: no recording and no clinical content are kept	Nothing. Do not retry.
400	<code>MISSING_FIELD</code> · <code>INVALID_IDENTIFIER</code> · <code>MALFORMED_JSON</code>	The request is wrong	Log it; a developer fixes it. Do not retry unchanged.
401	<code>INVALID_KEY</code>	Key missing or wrong	Log it and tell AIMS LAB. Do not retry.
413	<code>TOO_LARGE</code>	Over 1 MB	Log it and tell AIMS LAB
422	<code>SCHEMA_INVALID</code>	Stored in quarantine; named fields are wrong	Log it; a developer fixes it
429	<code>RATE_LIMITED</code>	Too many requests	Retry after the <code>Retry-After</code> delay
500 · 503	<code>SERVER_ERROR</code> · <code>UNAVAILABLE</code>	AIMS LAB's problem	Retry in the background (<code>SRS-CHB-05</code>)

6.3 Recorder to the AIMS LAB server (AIMS LAB internal)

Route	Credential	Purpose
POST /api/v2/device/enroll	Enrolment code	Exchange a one-time code for a device identity
POST /api/v2/grant/mint	Device token	Ask permission to record one consultation, and confirm it (\$5)
POST /api/v2/session/open	Device token	Open a session and submit hash-chain entry o
POST /api/v2/segment/authorize	Device token	Get a 300-second upload link to R2 for one piece
POST /api/v2/segment/commit	Device token	Declare a piece uploaded, with its SHA-256; the server reads it back and checks it
POST /api/v2/session/pause . /resume	Device token	Record a pause and its reason
POST /api/v2/session/close	Device token	Close the session and submit the end of the chain
GET /api/v2/session/{id}/receipts	Device token	Collect receipts that allow local deletion
POST /api/v2/heartbeat	Device token	Liveness, buffer depth, version, audio level figures
GET /api/v2/doctors	Device token	Doctor names for this clinic only
/api/v2/admin/*	Admin key	Clinics, doctors, enrolment codes, revocation, alerts

SRS-IF2-01 [M, I, AIMS] All traffic between the recorder and the AIMS LAB server shall use TLS 1.2 or later.

SRS-IF2-02 [M, T, AIMS] Separate credentials shall exist for devices, workers, administrators and CMED, and none shall work in place of another.

SRS-IF2-03 [M, T, AIMS] A device shall be able to read only its own clinic's data. GET /api/v2/doctors for another clinic shall return 403 .

6.4 Cloud storage — Cloudflare R2 — STO

SRS-STO-01 [M, I, AIMS] Two buckets shall be used: a **segment bucket** for audio pieces waiting to be merged, and a **copy bucket** for the lossless copy of each finished recording and its JSON (SRS-ARC-12).

SRS-STO-02 [M, I, AIMS] Each role shall have its own storage credential, allowed only what it needs. The recorder holds no storage key at all and uploads only through 300-second upload links. The API reads the segment bucket. The clean-up credential can delete from the segment bucket only. The cloud-copy credential can write to the copy bucket but not delete from it. The restore credential can only read. No credential can do everything.

SRS-STO-03 [M, I, AIMS] Object names in both buckets shall be built from the session ULID, the clinic code and the date, and shall never contain a patient identifier or name (SRS-UPL-04).

SRS-STO-04 [M, A, Joint] Before the first patient audio is stored, the ethics committee shall confirm that holding encrypted patient audio with a provider outside Bangladesh is acceptable under the study's approval (**OD-14**).

SRS-STO-05 [S, I, AIMS] Storage shall be reached only through the standard S3 interface, so the provider can be changed without changing the recorder (SRS-NFM-04).

SRS-STO-06 [M, D, AIMS] Every month, one day's recordings shall be restored from the copy bucket to a spare machine, decrypted, decoded and compared with the archive, and the result recorded. A backup is trusted only once it has been restored.

6.5 Workers on the UIU server

SRS-IF3-01 [M, I, AIMS] The archive worker and the cloud-copy worker shall make outbound connections only, and shall accept no incoming connection.

SRS-IF3-02 [M, T, AIMS] Workers shall claim work in batches, with a guard that prevents two workers from processing the same session at the same time.

7. Functional requirements

These are AIMS LAB's obligations. CMED builds none of them.

7.1 Audio capture — CAP

ID	Requirement	Pri	Ver	Owner
SRS-CAP-01	Audio shall be captured as WAV PCM, 44.1 kHz, mono, 16-bit — 88,200 bytes per second, 318 MB per hour (OD-12 proposes 48 kHz).	M	I	AIMS
SRS-CAP-02	Audio shall never be converted to a lossy format at any point — not before the archive, not in the archive, and not for the cloud copy.	M	I	AIMS
SRS-CAP-03	Capture shall begin within 500 ms of an accepted trigger.	M	A	AIMS
SRS-CAP-04	Capture shall run on its own thread and shall never be blocked by network, disk or screen work.	M	I	AIMS
SRS-CAP-05	If the microphone is lost mid-session, the recorder shall warn, try to recover, and keep everything captured so far.	M	T	AIMS
SRS-CAP-06	The recorder shall detect a session that stays silent — level below about –52 dBFS across two pieces in a row — and send one <code>silent_session</code> notification per session.	S	T	AIMS

On SRS-CAP-02 . Opus at 32 kbps would make recordings about twenty times smaller and was rejected. The archive is evidence. A lossy copy cannot be checked against what the microphone heard, which destroys the property the hash chain exists to prove. FLAC is allowed for the cloud copy because it is lossless: it decodes to exactly the same samples (**CON-07** , §8.5).

On SRS-CAP-06 . This is a level check on finished pieces — a microphone muted at its switch, unplugged, or set to the wrong input. It is different from the cutter's background-noise estimate in **SRS-SEG-03** , which only finds places to cut and raises no notification.

7.1a How the microphone is connected — CAP continued

Between the microphone and the WAV file sit several stages of optional processing, and every one is a place where a quiet speaker disappears. A consultation has two speakers at very different levels. The doctor is close to the device and speaks clearly. The patient is further away, often unwell or elderly, and speaks quietly. Any processing designed to "isolate the main speaker" — which is what conference hardware does — removes the patient.

Measurements on 128 archived sessions showed the pattern. In poor sessions, background noise sits at –87 to –97 dBFS with **8.7% of the audio at exact digital silence**, and the quiet speaker at –56 to –69 dBFS. In good sessions from the same clinics, background noise is –60 to –71 dBFS and the quiet speaker –36 to –42 dBFS. The loud speaker is at –18 to –21 dBFS in both. A microphone in a room with a fan cannot produce exact digital silence; only a noise gate can.

ID	Requirement	Pri	Ver	Owner
SRS- CAP-07	Capture shall use the device's own sample rate and channel count , with no resampling or channel mixing by the operating system.	M	T	AIMS
SRS- CAP-08	Capture shall use the low-latency Windows audio path (WASAPI) where available. The old MME path shall not be used.	M	I	AIMS
SRS- CAP-09	The microphone shall be chosen by name and audio path , never by number, because device numbers change between restarts and USB ports.	M	T	AIMS
SRS- CAP-10	Automatic gain control, noise suppression and gating in the device or in Windows shall be turned off wherever they can be.	M	I	AIMS
SRS- CAP-11	A machine shall not be put into service until a measurement shows less than 0.5% of the audio at digital silence over 60 seconds of an occupied room.	M	T	AIMS
SRS- CAP-12	That measurement shall record, per room: peak level, background noise, quiet-speaker level and loud-speaker level.	M	T	AIMS
SRS- CAP-13	Where the processing cannot be turned off, a microphone without built-in processing shall be used instead. A conference device whose processing cannot be disabled is not fit for recording evidence.	M	A	AIMS

Accepted range, taken from the good sessions in our own clinics:

Measure	Target	Reject
Peak (loud speaker)	−6 to −12 dBFS	above −3 dBFS (clipping)
Background noise	−60 to −75 dBFS	below −85 dBFS (gating)
Quiet speaker	above −45 dBFS	below −50 dBFS
Audio at digital silence	under 0.5%	0.5% or more

Turning up the gain does not help. The doctor is already recorded at the right level, and one archived session already clipped. More gain pushes the doctor into clipping without lifting the patient, because the patient is not too quiet — the patient is being gated out. Moving the microphone towards the patient helps by 6 to 10 dB and costs nothing: sound falls by 6 dB each time distance doubles, so moving closer to the patient takes level from the doctor, who has plenty, and gives it to the patient, who needs it.

7.1b Speech level monitoring — LVL

ID	Requirement	Pri	Ver	Owner
SRS-LVL-01	The recorder shall work out, for each finished piece, the background noise and the quiet and loud speech levels.	M	T	AIMS
SRS-LVL-02	These figures shall be sent to the AIMS LAB server for every session.	M	T	AIMS
SRS-LVL-03	When the quiet speaker falls below -50 dBFS while the loud speaker is above -25 dBFS, the on-screen control shall show a small prompt suggesting the patient speak a little louder.	S	D	AIMS
SRS-LVL-04	When both speakers are low, the prompt shall instead report a microphone problem, because a patient cannot fix that.	S	T	AIMS
SRS-LVL-05	When the loud speaker goes above -3 dBFS, the recorder shall warn that the recording is clipping.	S	T	AIMS
SRS-LVL-06	Level prompts shall appear at most once per session , shall never block the screen, and shall never need to be dismissed.	M	T	AIMS
SRS-LVL-07	The first piece of a session shall be exempt, while the background-noise estimate settles.	M	T	AIMS
SRS-LVL-08	Rooms that are below the accepted range day after day shall be reported to operations as a room or hardware fault , not handled by prompting the doctor.	M	I	AIMS

SRS-LVL-08 matters more than the prompt. A room that is too quiet every day needs its microphone moved or replaced. Asking a doctor to raise their voice thirty times a day hides a hardware fault behind a person.

7.2 Cutting audio into pieces — SEG

ID	Requirement	Pri	Ver	Owner
SRS-SEG-01	Pieces shall be 30–60 seconds long, cut at a quiet moment inside that window.	M	T	AIMS
SRS-SEG-02	If no quiet moment is found by 60 seconds, the cutter shall wait up to 15 seconds more, then cut at 75 seconds.	M	T	AIMS
SRS-SEG-03	Quiet moments shall be found using signal level (threshold 320, held for 3.0 s) against a background-noise estimate, together with zero-crossing rate.	M	I	AIMS
SRS-SEG-04	The cutter shall run on its own thread, so a slow disk cannot stall capture.	M	I	AIMS
SRS-SEG-05	Pieces shall be numbered from zero within a session, with no gaps.	M	T	AIMS
SRS-SEG-06	Quiet detection shall not depend on absolute loudness: the speech estimate shall be taken from the opening seconds of audio, not from a fixed number, so a quiet input still cuts in the right places.	M	T	AIMS
SRS-SEG-07	At least 80% of pieces in a commissioned room shall be 30–60 seconds long, measured over a clinic day.	M	A	AIMS

Why 30–60 seconds, not three minutes. A piece is the unit that is uploaded, retried and — at worst — lost. At three minutes, one failure risked three minutes of consultation and held back everything after it. At one minute, the risk is a minute, and a poor connection recovers in smaller steps. The server joins the pieces back into one file, so nothing is lost by cutting often.

7.3 The buffer on the PC — SPL

Nothing is kept on the doctor's PC. That is the rule; this section explains how it is kept.

Audio has to exist somewhere for a moment between being recorded and being sent. That moment is short. A piece is written to disk encrypted, uploaded, and deleted as soon as the server confirms it has a good copy. Normally this takes seconds. The buffer is a shock absorber, not a store: if the link drops for a few minutes, pieces wait and then go. If anything is still waiting after fifteen minutes, something is wrong, and the system says so.

ID	Requirement	Pri	Ver	Owner
SRS-SPL-01	Every finished piece shall be encrypted with AES-256-GCM, using its own key, before it is written to disk.	M	I	AIMS
SRS-SPL-02	The buffer's master key shall be protected by DPAPI.	M	I	AIMS
SRS-SPL-03	Each piece shall move through PENDING → COMMITTED → RECEIPTED → PURGED, with QUARANTINED as a separate branch.	M	T	AIMS
SRS-SPL-04	Each change of state shall be written to an append-only journal, flushed to disk, before the action it records.	M	T	AIMS
SRS-SPL-05	The buffer shall survive a sudden power loss without accepting a half-written piece as complete.	M	T	AIMS
SRS-SPL-06	The buffer shall be sized for a short interruption: 4 GB, about 13 hours of recording.	M	A	AIMS
SRS-SPL-07	The recorder shall raise an alert if the buffer is more than 25% full, or if the oldest undelivered piece is more than 15 minutes old. Either means delivery has stalled.	M	T	AIMS
SRS-SPL-08	Local audio shall be deleted as soon as a valid receipt arrives, with no waiting period. The receipt is the proof; waiting longer adds nothing.	M	T	AIMS
SRS-SPL-09	A QUARANTINED piece shall be delivered to the server before it is deleted, exactly like a verified one. It shall not be deleted while undelivered, and shall not be kept once delivered.	M	T	AIMS
SRS-SPL-13	No piece shall stay on the PC beyond one delivery cycle once the connection allows delivery. The age of the oldest undelivered piece shall be reported in every heartbeat.	M	T	AIMS
SRS-SPL-14	The buffer shall be empty whenever the server has been reachable and idle for one delivery cycle. A non-empty buffer on a connected machine is a fault and shall raise an alert.	M	T	AIMS

Audio held only in memory, stated honestly. Between capture and a piece being sealed, about **99 seconds** of audio exists only in memory — a 75-second capture buffer plus about 24 seconds in a queue. A sudden power loss during that window loses it. Cutting into pieces limits this; it does not remove it.

ID	Requirement	Pri	Ver	Owner
SRS-SPL-10	An intermediate save point shall be added within each piece, reducing the audio held only in memory without changing the cutting, chain or encryption design.	S	T	AIMS
SRS-SPL-11	The save-point interval shall be set from measured disk-flush times on the clinic PCs, not guessed (OD-09).	S	A	AIMS
SRS-SPL-12	The save point shall be proven across repeated forced shutdowns before it is described as working.	S	T	AIMS

7.4 Integrity — the hash chain — CHN

Each session carries a chain of signed entries: one when it opens, one for every piece, one for every pause and resume, and one when it closes. Every entry includes the fingerprint of the entry before it and is signed by the PC's private key. Removing, reordering or editing anything breaks the chain, and the break can be seen.

ID	Requirement	Pri	Ver	Owner
SRS-CHN-01	Each session shall carry an Ed25519-signed hash chain with entries of type <code>open</code> , <code>segment</code> , <code>pause</code> , <code>resume</code> and <code>close</code> .	M	T	AIMS
SRS-CHN-02	Each entry shall include the previous entry's fingerprint, so a missing or reordered entry is detectable.	M	T	AIMS
SRS-CHN-03	Fingerprints shall be SHA-256 with a label for each use — <code>aimscribe.chain.v2</code> , <code>aimscribe.receipt.v2</code> , and equivalents — so one kind cannot be passed off as another.	M	I	AIMS
SRS-CHN-04	Entries shall be signed with the device key registered at enrolment.	M	T	AIMS
SRS-CHN-05	The AIMS LAB server shall verify the whole chain when the session closes, and quarantine the session if it fails.	M	T	AIMS
SRS-CHN-06	Chain entries shall be written in one exact JSON form, so a signature can be reproduced byte for byte.	M	T	AIMS
SRS-CHN-07	A broken chain shall be repaired only by fixing the system code that broke it. Editing chain data by hand is prohibited.	M	I	AIMS

On SRS-CHN-07. A chain repaired by hand proves nothing. If a chain breaks, the fault is in the code that produced it, and patching the evidence by hand destroys the only reason the evidence exists.

7.5 Upload and checking on arrival — UPL

ID	Requirement	Pri	Ver	Owner
SRS-UPL-01	Each piece shall be uploaded to the R2 segment bucket through its own upload link, valid for 300 seconds.	M	T	AIMS
SRS-UPL-02	When a piece is declared uploaded, the AIMS LAB server shall read it back from R2 and recompute its fingerprint, comparing it with the declared SHA-256 in constant time.	M	T	AIMS
SRS-UPL-03	A mismatch on arrival shall be handled by §7.5a and shall raise an alert.	M	T	AIMS
SRS-UPL-04	Object names shall be built from the session ULID and a prefix, never from a patient identifier.	M	I	AIMS
SRS-UPL-05	Uploads shall retry with a growing delay and shall never block capture or cutting.	M	T	AIMS
SRS-UPL-06	The recorder shall keep recording without a connection, holding pieces in the buffer and uploading them when the connection returns.	M	D	AIMS

On SRS-UPL-02 , and its cost. Because every piece is read back, the UIU server downloads the full audio volume from R2 — about 10 Mbit/s at the busiest time — and §10 sizes its internet line for that. The alternative, trusting the recorder's own fingerprint, would make the whole chain decorative: a damaged upload would be recorded as verified.

On SRS-UPL-04 . Object names appear in access logs, dashboards and error reports. A patient identifier in a name would appear in all three.

7.5a When a piece fails its check — REC

The first thing to know: a failed check on the server almost never means damaged audio.

The recorder decrypts and re-checks every piece against its recorded fingerprint *immediately before each upload*. If the file on the PC is damaged, it is caught there. So a piece that reaches the server was verified good moments earlier. And most failures are not about the audio at all.

7.5a.1 Every way a session can be quarantined

#	Cause	Where	Is the audio damaged?	Likely reason
1	local_segment_unreadable	Recorder, before upload	Possibly	Disk damage while waiting; a half-written file after power loss; antivirus changing the file; a DPAPI key that no longer works after a Windows profile rebuild
2	Fingerprint mismatch on arrival	Server, on commit	No — checked just before upload	Upload cut short; upload link expired mid-transfer; a retried upload leaving a partial object; a storage-side fault
3	Chain entry rejected	Server, on commit	No	An entry arriving before the one it follows; a lost pause or resume entry; a device key that no longer matches
4	Same piece number with a different fingerprint	Server, on commit	No	A commit that succeeded but whose reply was lost, retried with different bytes; two copies of the recorder using one buffer
5	Whole-chain check fails	Server, on close	No	Any gap, break or bad signature built up across the session

Only row 1 puts the recording itself at risk, and row 1 is caught on the PC. Rows 2 to 5 describe a consultation whose audio is complete and correct, held back by a fault in the paperwork around it.

ID	Requirement	Pri	Ver	Owner
SRS-REC-01	When the server rejects a piece, the recorder shall re-check its local copy and, if it passes, treat the failure as recoverable.	M	T	AIMS
SRS-REC-02	If the local copy is good, the piece shall be uploaded again to a new object name and committed again. The suspect object shall not be overwritten; it is evidence of the failure.	M	T	AIMS
SRS-REC-03	Retries shall be limited to three, with growing delay. A rejected piece shall not be retried on every cycle.	M	T	AIMS
SRS-REC-04	If the local copy is not good, the recorder shall stop retrying, mark the piece unverifiable, and upload it under SRS-REC-07 anyway. It shall not keep it.	M	T	AIMS
SRS-REC-05	The server shall quarantine the piece , and the whole session only once retries are exhausted.	M	T	AIMS
SRS-REC-06	Quarantine shall be resolved by an automatic process on the server. It shall not require an administrator, and never any action on a clinic PC.	M	T	AIMS
SRS-REC-07	A piece that cannot be verified shall still be uploaded, to a separate quarantine/ area, outside the chain and marked unverifiable. It shall not enter the evidence archive, and it shall not remain on the PC.	M	T	AIMS
SRS-REC-08	Every heartbeat shall report the number and size of pieces stuck on the PC, so a PC holding audio is visible centrally, not only in its own log.	M	T	AIMS
SRS-REC-09	An operator view shall list sessions stuck in quarantine across all clinics, with clinic, room, age and size.	M	D	AIMS

Evidence and data are different goals. The chain protects evidence. A consultation whose chain broke still holds clinical speech worth keeping — and a prescription cannot be drafted from a recording that never arrived. Keeping the two goals apart lets both hold: the archive stays strictly verified, and the unverifiable recording arrives anyway, with its fault recorded against it.

7.5a.2 Deletion follows custody, not archiving

In September 2026, audio was found still sitting on clinic PCs days after it had been uploaded and verified. Delivery had worked. Deletion had not, for two reasons, both since removed:

1. **Deletion waited for the archive.** A receipt was issued only after the archive worker had merged and filed the session, so a verified piece could wait on a PC for a process elsewhere to run.
2. **A 24-hour wait followed the receipt.** Clinic PCs are switched off after clinic hours, so the wait usually expired while the PC was off, and deletion slipped day after day.

SRS-REC-15 [M, T, AIMS] A receipt allowing deletion shall be issued as soon as the AIMS LAB server holds a verified copy of the piece in R2. It shall not wait for archiving, which is an internal AIMS LAB step and shall not decide what stays on a clinic PC.

SRS-REC-16 [M, T, AIMS] Deletion shall follow the receipt immediately. No waiting period shall come between a verified receipt and removal of the local copy.

SRS-REC-17 [M, T, AIMS] The recorder shall clear anything deletable at start-up, before it begins recording, so a PC switched on in the morning clears the previous day within seconds.

SRS-REC-18 [M, T, AIMS] The recorder shall try a final delivery and clean-up at shutdown, within a time limit, so a laptop closed straight after the last patient leaves as little behind as possible.

7.5a.3 Nothing waits on the PC for a person

Nothing waits on a clinic PC for a decision, because nobody there will make one. The doctor is seeing patients. A file waiting for review on a laptop in Ershadnagar is a file nobody will ever review.

SRS-REC-10 [M, T, AIMS] Every finished piece shall reach the server, without exception and without human involvement. Its check decides *where* it lands — the evidence archive or quarantine — never *whether* it is sent.

SRS-REC-11 [M, T, AIMS] A scheduled sweep shall deliver anything not yet sent — pending, failed, quarantined, or left by a crash. It shall run separately from normal upload, and shall not skip a piece because an earlier one failed.

SRS-REC-12 [M, T, AIMS] Local copies shall be deleted once the server confirms it has them — a receipt for verified audio, or an intake confirmation for quarantined audio. Deletion follows custody, not the check result, and not archiving.

SRS-REC-13 [M, I, AIMS] No normal or exceptional procedure shall require a person to open, copy, inspect or delete a file on a clinic PC.

SRS-REC-14 [M, T, AIMS] Quarantined material shall be reviewed by a server process that reaches its own conclusion: it re-checks the piece against the chain, admits it to the archive if it now matches, or files it permanently as unverifiable. A person is told the result; nobody is waited for.

7.6 Sessions and the handover between patients — SES

ID	Requirement	Pri	Ver	Owner
SRS-SES-01	Sessions shall be identified by a ULID created by the AIMS LAB server.	M	I	AIMS
SRS-SES-02	A new trigger, when the gate allows it, shall close the current session and open the next with no break in capture.	M	D	AIMS
SRS-SES-03	The new session's start time shall become the previous session's end time, so consultations follow each other with no gap or overlap.	M	T	AIMS
SRS-SES-04	Closing one session and opening the next shall run on separate threads, so the handover costs no recording time.	M	D	AIMS
SRS-SES-05	The archived files shall be named PatientID_DoctorID_HospitalID_HHMMSS_HHMMSS_YYYYMMDD — patient, doctor, AIMS LAB clinic code, start time, end time, date — for example P0012345_DR0042_HOSP003_101432_102847_20260913 .	M	I	AIMS
SRS-SES-06	Every close shall record a reason, telling normal closes apart from abnormal ones.	M	T	AIMS
SRS-SES-07	superseded_by_new_patient shall be a normal close and shall not need a reason from the doctor.	M	T	AIMS
SRS-SES-08	Abnormal closes shall raise an integrity alert, keyed on the reason.	M	T	AIMS
SRS-SES-09	A crash mid-session shall leave a recoverable session. On restart, the recorder shall close it with an explicit abnormal reason rather than discard it.	M	T	AIMS

On SRS-SES-05 . Times are written HHMMSS because Windows does not allow : in file names. The clinic part is AIMS LAB's code (HOSP003), not CMED's identifier.

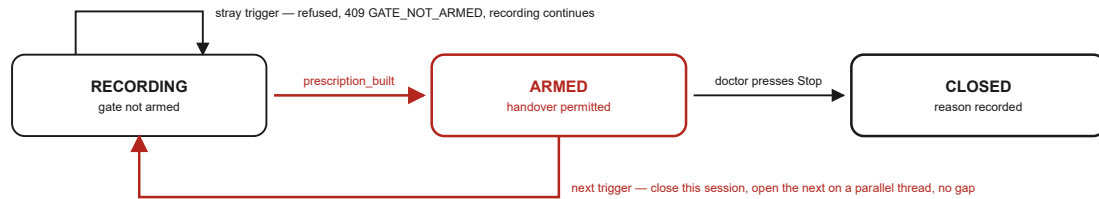
On SRS-SES-07 . Asking the doctor for a reason at every automatic handover would mean thirty forms a day for the most ordinary event in the clinic. The reason form in §7.8 applies only to deliberate stops.

7.7 The consultation gate — GAT

The problem. Mid-consultation, a doctor clicks another patient in the waiting list — to check something, or by mistake. If every trigger ended the current recording, the consultation would be cut in half, with no warning and no way to know.

The solution. A new trigger does not end the current session until that session's prescription has been built.

A new patient cannot end the current recording until that consultation's prescription has been built.



Residual risk: a patient who leaves before a prescription is built never arms the gate, and the session stays open until the doctor stops it. That is why the overlay's Stop button is always visible.

Figure 9. The consultation gate. A session starts unarmed, and a stray trigger is refused while the recording continues. Only `prescription_built` arms the gate; after that, the next trigger hands over to the next patient with no gap.

ID	Requirement	Pri	Ver	Owner
SRS-GAT-01	A session shall start unarmed.	M	T	AIMS
SRS-GAT-02	Only <code>prescription_built</code> for the open session's patient and session shall arm the gate.	M	T	AIMS
SRS-GAT-03	A trigger arriving while unarmed shall be refused with <code>409 GATE_NOT_ARMED</code> ; the current recording shall continue undisturbed.	M	T	AIMS
SRS-GAT-04	A trigger arriving while armed shall perform the handover of <code>SRS-SES-02</code> .	M	T	AIMS
SRS-GAT-05	The Stop button shall close a session whether or not the gate is armed, subject to §7.8.	M	T	AIMS
SRS-GAT-06	Every refused trigger shall be logged with patient, doctor and time, and shown in operational reports.	M	T	AIMS
SRS-GAT-07	Gate state shall be included in <code>status</code> events, so CMED may show it if it wishes.	S	T	AIMS

The remaining risk. Sometimes a patient leaves before a prescription is built: sent for a test, referred elsewhere, or simply gone. The gate then never arms, and the recording stays open until the doctor presses Stop or opens the next patient's prescription. This is why the Stop button is always visible. **OD-04** asks what share of consultations end without a prescription, because that decides whether a time limit is also needed.

7.8 The on-screen Stop and Pause control — `UIX`

Before this control, a doctor who wanted to pause had to find the Windows tray, locate AIMScribe, right-click and choose from a menu — during a consultation, with a patient watching. A control nobody can use is a control nobody uses.

The control. When recording starts, a small always-on-top window appears at the top right of the screen, in front of the browser. It is fixed in place and small enough to cover nothing clinically important.

Control	Shape	Colour	Action
Stop	Circle	Red	Ends the session — after a reason is given
Pause	Rectangle	Blue	Pauses recording — after a reason is given

ID	Requirement	Pri	Ver	Owner
SRS- UIX-01	The control shall appear automatically when a session starts, and disappear when it closes.	M	D	AIMS
SRS- UIX-02	It shall stay on top, fixed at the top right, and shall not need to be moved.	M	D	AIMS
SRS- UIX-03	It shall have exactly two buttons: a red circular Stop and a blue rectangular Pause.	M	I	AIMS
SRS- UIX-04	Pressing Pause shall show a list of reasons and a free-text comment box.	M	D	AIMS
SRS- UIX-05	The pause shall take effect only after a reason is chosen or typed and confirmed.	M	T	AIMS
SRS- UIX-06	Pressing Stop shall show the same form, and the session shall close only after a reason is chosen or typed and confirmed.	M	T	AIMS
SRS- UIX-07	The reason form shall not close without either a reason being given or the action being cancelled.	M	T	AIMS
SRS- UIX-08	On Stop, the microphone shall cut the instant the button is pressed. The reason decides how the session is <i>filed</i> , not whether the doctor stops being recorded.	M	T	AIMS
SRS- UIX-09	Reasons shall be recorded in the chain, as signed <code>pause</code> , <code>resume</code> and <code>close</code> entries.	M	T	AIMS
SRS- UIX-10	A pause longer than a set duration shall be flagged for review.	S	T	AIMS
SRS- UIX-11	The control shall work from the keyboard and be readable at 125% and 150% display scaling.	S	D	AIMS
SRS- UIX-12	The control shall require nothing from CMED.	M	I	AIMS
SRS- UIX-13	While a recording waits for confirmation (SRS-CNF-07), the control shall show "confirming" in plain words, without an error or a sound.	M	D	AIMS
SRS- UIX-14	When a recording is stopped because API 2 records no consent (SRS-CNF-11), the control shall say in plain words that this consultation is not being recorded, then close. It shall ask the doctor for nothing.	M	D	AIMS

On SRS-UIX-08 . A doctor cannot stop a recording without a reason, so a vague stop cannot quietly damage data collection. But if the doctor pressed Stop because the patient just objected to being recorded, the microphone must stop *now*, not after a form. So the press cuts the microphone at once, and the form decides how the session is filed. The session stays open and unfiled until the form is completed, but the patient's objection is honoured instantly.

7.9 The AIMS LAB server — BKD

ID	Requirement	Pri	Ver	Owner
SRS-BKD-01	Consent shall be enforced separately in three places: the grant, which requires <code>consent_obtained: true</code> in the matching API 2; the recorder, which keeps no audio without a grant; and a database constraint, which refuses a recording linked to a visit without consent.	M	T	AIMS
SRS-BKD-02	An <code>audit_log</code> shall record every security-relevant event. Rows shall never be edited or deleted, enforced by a database trigger rather than by convention.	M	T	AIMS
SRS-BKD-03	Every write shall be safe to retry; a repeated commit shall not create a duplicate row.	M	T	AIMS
SRS-BKD-04	Identifiers shall be checked against <code>^[A-Za-z0-9_-]{1,64}\$</code> wherever they enter the system.	M	T	AIMS
SRS-BKD-05	Error replies shall not reveal which of several causes occurred where that would help an attacker.	M	I	AIMS
SRS-BKD-06	The server shall provide a health endpoint reporting whether the databases and R2 can be reached.	M	D	AIMS

On SRS-BKD-04 . Identifiers become folder and file names on the archive. This one pattern is what stops a stray `../` from escaping the archive folder.

7.10 The archive and the cloud copy — ARC

ID	Requirement	Pri	Ver	Owner
SRS-ARC-01	The archive worker shall archive only sessions whose chain has been verified and whose recording is confirmed (§5.6).	M	T	AIMS
SRS-ARC-02	It shall join the pieces, in order, into one WAV file per consultation.	M	T	AIMS
SRS-ARC-03	It shall check the joined file against the hash chain before reporting completion.	M	T	AIMS
SRS-ARC-04	It shall file each recording under <code>/archive/<hospital_id>/<doctor_id>/<YYYY-MM-DD>/</code> , using AIMS LAB's clinic code.	M	I	AIMS
SRS-ARC-05	<i>Withdrawn in 3.0.</i> Receipts are now issued on custody, not after archiving — see SRS-REC-15 .	—	—	—
SRS-ARC-06	It shall check for work at a fixed interval, in batches of limited size, and shall wait and retry calmly when a database or R2 is unreachable.	M	T	AIMS
SRS-ARC-07	The archive array shall be backed up — by the copy bucket (SRS-ARC-09) — and the backup restore-tested (SRS-STO-06).	M	D	AIMS
SRS-ARC-08	Pieces in R2 shall not be deleted as soon as a session is archived. Each finished consultation shall first be kept in the cloud as one merged, losslessly compressed copy.	M	T	AIMS
SRS-ARC-09	The steps shall run in this order, and each shall succeed before the next begins: (1) the merged WAV and its JSON are written to the UIU archive and the WAV checked against the hash chain; (2) the WAV is compressed to FLAC; (3) the FLAC is decoded and confirmed to contain exactly the samples of the archived WAV; (4) the FLAC and the JSON are encrypted and uploaded to the copy bucket; (5) they are downloaded again and confirmed complete; (6) the cloud copy is recorded in the recordings database; (7) only then are the pieces deleted from the segment bucket.	M	T	AIMS
SRS-ARC-10	If any step fails, the pieces shall be kept and the step retried. Repeated failure shall raise an alert. No step shall ever be skipped to free space.	M	T	AIMS
SRS-ARC-11	The cloud copy shall be lossless (FLAC). A lossy format shall not be used, because a lossy copy cannot rebuild the archive or be checked against the hash chain.	M	I	AIMS
SRS-ARC-12	Pieces and merged copies shall be kept in separate buckets. The copy bucket shall be locked against deletion; the segment bucket shall not be, so clean-up can run. The credential that deletes pieces shall have no access to the copy bucket.	M	I	AIMS
SRS-ARC-13	When a prescription arrives after the recording is archived, the JSON file beside it shall be rewritten, and the new JSON uploaded to the copy bucket as a new object. Earlier versions are kept.	M	T	AIMS

ID	Requirement	Pri	Ver	Owner
SRS- ARC- 14	Recordings shall normally be archived within five minutes of closing, and their cloud copy made the same night.	S	A	AIMS
SRS- ARC- 15	Pieces downloaded to the UIU working disks for checking shall be deleted once the recording is archived and verified.	M	T	AIMS
SRS- ARC- 16	An unconfirmed recording (SRS-CNF-08) shall not be archived or copied until it is confirmed. If it is deleted under SRS-CNF-09 , its pieces shall be deleted from R2 and the deletion recorded in the audit log.	M	T	AIMS

Why the order matters. The pieces are deleted only after two things are proven: the UIU server holds the recording and it matches the chain, and the cloud holds a lossless copy that decodes to the same samples. Until step 7, at least two complete copies of the audio exist, so a failure part way through costs time, never audio.

8. Data requirements

8.1 What crosses between CMED and AIMS LAB

CMED → AIMS LAB

Data	Sent in	Kind of data	Kept?
The five fields	API 1, API 2, API 3	Identifiers and times	Yes — file name and both databases
<code>session_id</code>	<code>prescription_built</code>	Opaque identifier AIMS LAB created	Used to arm the gate only
Name, sex, age, phone, address	API 2	Personal data	Yes — clinical database and JSON file only
Paramedic measurements and notes	API 2	Clinical	Yes
Previous prescription	API 2	Clinical	Yes
Prescription, diagnoses, tests, advice, follow-up	API 3 (Channel B)	Clinical	Yes
<code>consent_obtained</code> , <code>consent_method</code> , <code>consent_recorded_at</code>	API 2	Consent record	Yes — clinical database, JSON file and audit log. Without consent, nothing else is kept.

AIMS LAB → CMED

Data	Notes
<code>status</code> , <code>code</code> , <code>message</code>	The outcome of each command or request
<code>session_id</code> , <code>record_id</code>	Opaque identifiers
Recording, gate and confirmation state	For an optional indicator
Warnings	Disk, microphone, connection
Audio	Never. In no form, at no time.
Transcripts or AI output	Never through these interfaces

SRS-DAT-01 [M, I, AIMS] Audio shall never be sent to CMED, in any form, through any interface in this document.

SRS-DAT-02 [M, T, AIMS] A patient's name shall never appear in a hash-chain entry, a file name, an archive path, a cloud object name or a log line. It is stored only in the clinical database (§8.7) and in the JSON file beside the recording (§8.8), where access is separately controlled.

SRS-DAT-03 [M, I, Joint] Clinical content crosses in one direction only: from CMED to AIMS LAB. Nothing clinical is ever sent back.

What holding patient information changes. Once the archive holds patient information, it is no longer a store of unlabelled audio.

While only audio was held	Now that patient information is held
A folder name revealed nothing	Each recording sits beside a file naming the patient
Consent covered audio recording	Consent must also cover collecting the clinical record
Losing a disk exposed unlabelled audio	Losing a disk exposes an identified clinical record

SRS-DAT-15 [M, I, Joint] The consent text given to patients shall be revised before any clinical record is collected, and the revision dated. Consent for audio recording alone does not cover collecting the record.

SRS-DAT-16 [M, I, AIMS] The archive and the clinical database shall be encrypted at rest. Access to the clinical database shall be granted separately from access to the recordings catalogue: a role that can search recordings shall not thereby be able to read prescriptions.

8.2 Identifiers and file names

SRS-DAT-04 [M, T, Joint] Every identifier shall match `^[A-Za-z0-9_-]{1,64}$`.

SRS-DAT-05 [M, I, CMED] Identifiers shall stay the same over time. If a patient's `patient_id` changes, their recordings become two unrelated sets that cannot be joined again (**OD-02**).

SRS-DAT-06 [M, I, CMED] Identifiers shall not contain personal data. A patient identifier containing a name or national ID number would put that data into file names, logs and dashboards.

The recording's name. Every recording has one name, used in three places. This name is how everything about a visit is found.

P0012345_DR0042_HOSP003_101432_102847_20260913

Part	Meaning
P0012345	Patient ID, from CMED
DR0042	Doctor ID, from CMED
HOSP003	AIMS LAB clinic code, from the PC's enrolment
101432	Start time, HHMMSS
102847	End time, HHMMSS
20260913	Date, YYYYMMDD

Where	How the name is used
UIU archive	<code>/archive/HOSP003/DR0042/2026-09-13/<name>.wav</code> and <code><name>.json</code>
<code>aims_recordings.sessions</code>	Column <code>file_stem</code>
<code>aims_clinical.encounters</code>	Column <code>file_stem</code>

8.3 How long data is kept

Data	Where	Kept until
Encrypted pieces	Doctor's PC	The receipt — then deleted at once
Pieces	R2 segment bucket	The archive and the cloud copy are both verified (SRS-ARC-09) — then deleted
Pieces of an unconfirmed recording	R2 segment bucket	Confirmed, or 24 hours without confirmation — then deleted (SRS-CNF-09)
Working copies	UIU working disks	The recording is archived (SRS-ARC-15)
WAV and JSON	UIU archive	OD-06 — not yet decided
FLAC and JSON copy	R2 copy bucket, locked	OD-06 ; the lock period is OD-20
Patient information waiting to be matched	aims_recordings	Five minutes (SRS-CNF-03); the information itself is kept as a clinical record
Notice of a consultation without consent	aims_recordings	24 hours — the five fields and time only — then deleted (SRS-CNF-12)
Clinical records	aims_clinical	OD-17 — decided with the ethics committee
Session details	aims_recordings	Indefinitely
Audit logs	Both databases	Indefinitely; never edited
Database backups	R2, encrypted	Nightly; the retention period follows OD-17
Recorder logs	Doctor's PC	30 days

SRS-DAT-07 [M, A, AIMS] A retention period for audio shall be decided before hardware and storage are bought. At about 11 TB a year on the archive and 5 TB a year in the cloud, retention is the largest cost in the system (**OD-06**).

8.4 Protecting data

SRS-DAT-08 [M, I, AIMS] Audio shall be encrypted at rest at every stage: AES-256-GCM in the buffer on the PC, the provider's encryption in the segment bucket, and encrypted disks on the UIU server. The FLAC copy and its JSON shall also be encrypted on the UIU server before upload, with a key that never leaves UIU, so the storage provider holds data it cannot read.

SRS-DAT-09 [M, I, AIMS] Audio and clinical data shall be encrypted in transit at every stage, with TLS 1.2 or later.

SRS-DAT-10 [M, I, AIMS] No secret shall appear in either public source repository. Configuration files holding secrets shall be excluded from version control, and enrolment code sheets and the CMED API key shall be treated as credentials.

8.5 The original, the backup, and the transcription copy

Improving audio for speech recognition and keeping it as evidence make opposite demands on a file. They are not reconciled by compromise, but by keeping separate copies with separate rules.

	Archive WAV — the original	FLAC in the cloud — the backup	Transcription copy — derived
Where	UIU archive	R2 copy bucket	Wherever the ASR pipeline runs
Processing	None, ever	None — lossless compression only	Whatever helps recognition
Checked against the chain	Yes	Yes, after decoding	No
Can rebuild the original	It is the original	Yes	No
Kept	Per retention policy	Per retention policy	Disposable; can be recreated
Format	WAV PCM, capture rate, mono	FLAC, same rate and samples	16 kHz mono

SRS-DAT-11 [M, I, AIMS] The archived recording shall be bit-identical to the captured audio. No filtering, level change, noise reduction or re-encoding shall ever be applied to it.

SRS-DAT-12 [M, T, AIMS] Any copy prepared for speech recognition shall be made from the archived original, stored separately, and shall never replace it or be treated as evidence.

SRS-DAT-17 [M, T, AIMS] The FLAC copy in the cloud is a backup of the original, not a transcription copy. Because FLAC is lossless it decodes to samples identical to the archived WAV, so it may be used to rebuild the archive. It shall never be filtered, levelled or re-encoded.

SRS-DAT-13 [M, I, AIMS] The transcription processing and its settings shall be versioned and recorded, so every transcript can be traced to exactly how its audio was prepared.

SRS-DAT-14 [S, A, AIMS] Transcription processing shall avoid aggressive noise reduction and fast-acting compression, both of which add artefacts that make speech recognisers invent words. Room noise measured in our clinics sits at 100–300 Hz, so a gentle 80 Hz high-pass filter and slow level normalisation give most of the benefit.

8.6 Receiving patient information and prescriptions — CRI

The clinical record arrives at two moments, both server to server on Channel B (§6.2). The Channel A messages carry no clinical data.

Moment	What CMED's server sends	Why then
Patient opened	API 2: consent, demographics, paramedic measurements and notes, and the previous prescription for a returning patient	The doctor needs the history at the start. It also confirms the recording (§5.6).
Prescription built	API 3: prescription, diagnoses, tests, advice, notes	These do not exist until the prescription is built — the same moment the recording is armed

SRS-CRI-01 [M, T, CMED] Patient information shall be sent by CMED's server to the AIMS LAB server at the moment the patient is opened, carrying the five fields, the consent record, and a `demographics` object with the details recorded before the patient entered the room.

SRS-CRI-02 [M, T, CMED] For a patient who has visited before, the patient information shall also carry `previous_visit` : the complete most recent prescription, with its date.

SRS-CRI-03 [M, T, CMED] When the prescription is built, CMED's server shall send its contents, diagnoses, notes and tests ordered, carrying the same five fields as the trigger, including the consultation's original `start_time` .

SRS-CRI-04 [M, T, AIMS] Receiving clinical data shall never hold up recording. A record that is missing, late, malformed or rejected shall be logged and reconciled later; the recording continues regardless.

SRS-CRI-05 [M, T, AIMS] Each consultation's clinical record shall be written as a JSON file beside its audio, in the same folder and under the same name with `.json` (§8.8). It shall be written when the recording is archived, and rewritten if the prescription arrives later (`SRS-ARC-13`).

SRS-CRI-06 [M, T, AIMS] The record shall also be loaded into the clinical database (§8.7) field by field. The file travels with the audio; the database is the index. Neither is built from the other at query time.

SRS-CRI-07 [M, T, AIMS] Each record shall be checked against a published schema on arrival. A record that fails shall be stored unchanged in quarantine and reported — never silently discarded, and never partly loaded.

SRS-CRI-08 [M, I, AIMS] The patient's name shall be taken from the record and stored only in the clinical database and the JSON file. It shall not enter the file name, the folder path, the chain or the recordings database.

SRS-CRI-09 [M, T, AIMS] Receiving shall be idempotent. A record delivered twice shall produce one row, not two.

SRS-CRI-10 [M, T, AIMS] Records and recordings shall be reconciled every night. A recording with no record and a record with no recording shall both be reported; they are different faults with different causes.

SRS-CRI-11 [S, I, Joint] The fields required at each moment shall be agreed in writing and versioned. A field added later shall not invalidate records already stored under an earlier version.

SRS-CRI-12 [M, I, AIMS] Records shall be encrypted at rest, and shall not be sent to any downstream service that does not need them.

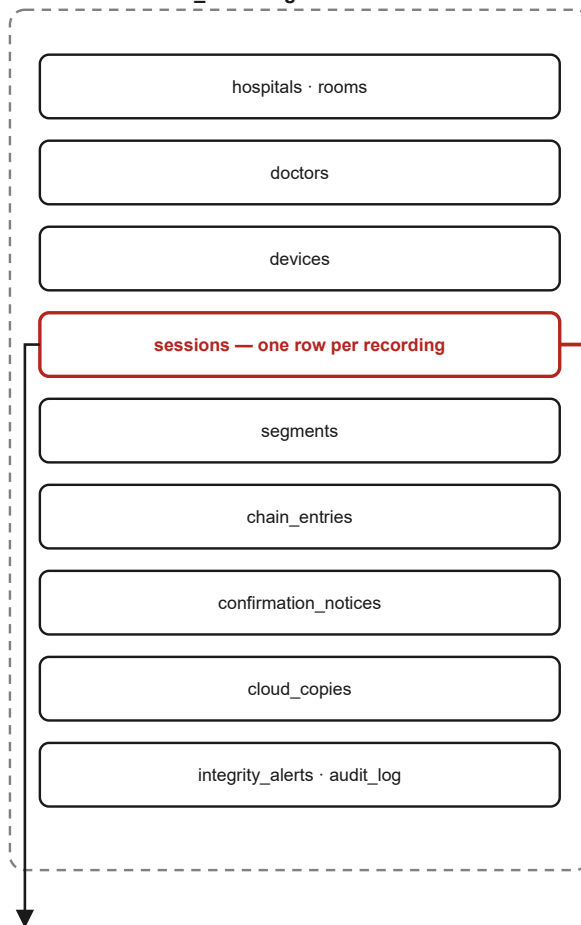
Reconciliation is a safety net. Most records are linked to their recordings at the moment of confirmation (`SRS-CNF-10`). The nightly check finds what that missed — a prescription that never arrived, or a recording that stayed unconfirmed.

8.7 The two databases — DBA

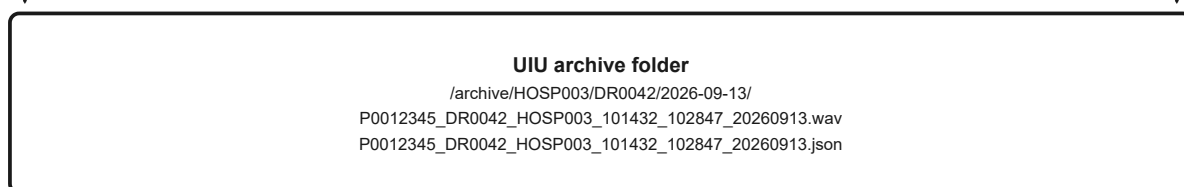
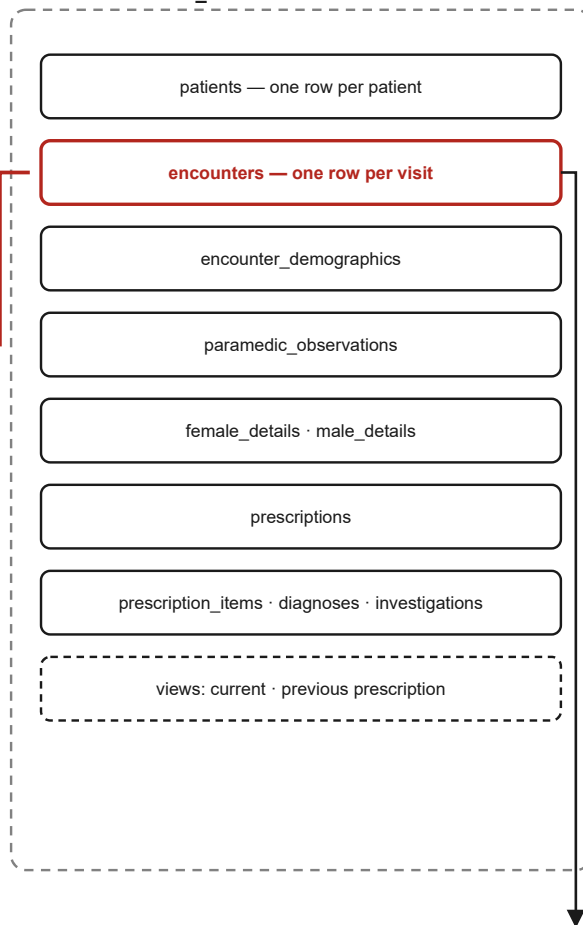
Two databases, one shared file name

They share only patient_id and the recording file name.

DATABASE 1 · aims_recordings



DATABASE 2 · aims_clinical



sessions points to the .wav; encounters points to the .json.

The accent line is the link: patient_id plus the shared file name.

Figure 10. Two databases, one shared file name. `aims_recordings` tracks recordings; `aims_clinical` holds patient information. They share only `patient_id` and the recording's file name, which is also the name of the WAV and JSON files on the UIU archive.

	aims_recordings	aims_clinical
Holds	Every recording, its pieces, its integrity checks, the devices and clinics	Patients, visits, paramedic measurements, prescriptions, diagnoses
Grows by	About 560 recordings a day	About 560 visits a day
Used by	The dashboard, the recorder API, the workers	The prescription pipeline and research
Contains patient names	No	Yes
Who can read it	Operations and engineering	A smaller, named clinical group (OD-19)

Separating them costs a join made in application code rather than by the database. That cost is accepted: it lets an engineer run the whole fleet without ever seeing a patient's name.

SRS-DBA-20 [M, I, AIMS] The recordings database and the clinical database shall be separate databases with separate credentials. No application role shall be able to write to both.

SRS-DBA-21 [M, T, AIMS] `patient_id` and the recording's file name (`file_stem`) shall be the only fields the two databases share. Reconciliation across them shall be done by a dedicated read-only role.

SRS-DBA-22b [M, I, AIMS] The dashboard shall read the recordings database only, and shall need no credential for the clinical database.

8.7.1 aims_recordings — what was recorded, where it is, and can it be trusted

Table	One row per	Main columns
hospitals	Clinic	hospital_id (never changes), name , operator , CMED's clinic identifier
rooms	Consulting room	room_id , hospital_id , label
doctors	Doctor at a clinic	doctor_id , hospital_id , full_name , active
devices	Registered PC	device_id , hospital_id , room_id , public_key , last_seen_at , revoked_at
enrollment_tokens	Enrolment code	Hash only, expires_at , used_at
sessions	Recording	session_id , file_stem , hospital_id , doctor_id , session_date , patient_id , room_id , device_id , started_at , ended_at , status , confirmation , close_reason , chain_verified_at , quarantine_reason , archive_path , archive_sha256
segments	Piece	session_id , seq_no , object_key , sha256 , bytes , state
chain_entries	Signed chain entry	session_id , entry_no , prev_hash , entry_hash , signature
confirmation_notices	API 2 waiting to be matched	The five fields, received_at , expires_at , claimed_by_session
cloud_copies	FLAC copy in the cloud	session_id , object_key , codec , bytes , verified_at , pieces_deleted_at
integrity_alerts	Problem needing attention	alert_type , severity , session_id , resolved_at
audit_log	Security-relevant event	Append-only; rows can never be edited or deleted

SRS-DBA-01 [M, I, AIMS] The recordings database shall model the tables above, with foreign keys enforced by the database rather than implied.

SRS-DBA-02 [M, T, AIMS] The hierarchy clinic → doctor → date → patient shall be served by an index on (hospital_id, doctor_id, session_date, patient_id) .

SRS-DBA-03 [M, T, AIMS] Every session row shall carry its archive file name and path, so a recording can be found by a query alone, without searching folders.

SRS-DBA-04 [M, T, AIMS] Each session shall record its close reason, whether its chain verified, whether it was quarantined and why, whether the doctor stopped it early, and its confirmation state.

SRS-DBA-22 [M, T, AIMS] There shall be one authoritative catalogue of recordings. The archive worker shall write each archived file's name, path, size and fingerprint back into it, so a single query answers where a recording is and whether it is verified.

SRS-DBA-23 [M, I, AIMS] The database shall never suspend or scale down to zero. Waking a sleeping database is indistinguishable, to a user, from a slow search.

SRS-DBA-24 [S, T, AIMS] Searching for part of a file name shall use a trigram index (pg_trgm), so the search stays fast as recordings grow from hundreds to hundreds of thousands.

SRS-DBA-25 [M, T, AIMS] The parts of the file name — patient, doctor, clinic, start, end, date — shall each be stored in their own column, and never pulled out of the file name at query time. The file name is for people; the columns are for queries.

8.7.2 aims_clinical — who the patient was, what was measured, what was prescribed

Table	One row per	Holds
patients	Patient	patient_id (primary key), sex , full_name , date_of_birth , phone , address
encounters	Visit	encounter_id , patient_id , doctor_id , hospital_id , start_time , date , file_stem , session_id , source , consent_obtained , consent_method , consent_recorded_at
encounter_demographics	Visit	Details as CMED sent them that day, so older visits still show what was true then
paramedic_observations	Visit	Blood pressure, pulse, temperature, oxygen, weight, height, notes
female_details	Visit, female patients only	Fields that apply only to female patients (OD-15)
male_details	Visit, male patients only	Fields that apply only to male patients (OD-15)
prescriptions	Prescription version	encounter_id , version , issued_at , advice , follow_up_date , notes , the original JSON
prescription_items	Medicine	line_no , drug , dose , frequency , duration , instructions
diagnoses	Diagnosis	line_no , text , optional code
investigations	Test ordered	line_no , name

source says where a visit came from: live for a consultation that was recorded, or previous_visit for an older prescription CMED sent in API 2.

Female and male patients: one database, the right fields for each. The fields needed differ by sex. Rather than two databases — which would turn "how many patients at Dholpur today" into two queries, and let any report that forgets one side under-count silently — each gets its own table, and the database itself enforces which table a patient may use:

```
-- A patient's sex is part of a unique key...
ALTER TABLE patients ADD UNIQUE (patient_id, sex);

-- ...so a female-only table can require it.
CREATE TABLE female_details (
    encounter_id UUID PRIMARY KEY REFERENCES encounters(encounter_id),
    patient_id    TEXT NOT NULL,
    sex           TEXT NOT NULL DEFAULT 'female' CHECK (sex = 'female'),
    FOREIGN KEY (patient_id, sex) REFERENCES patients (patient_id, sex)
    -- plus the female-only fields, once agreed with clinicians (OD-15)
);
```

A male patient's visit **cannot** be saved into `female_details` : the database refuses it, whatever the application does. `male_details` works the same way.

SRS-DBA-05 [M, I, AIMS] Patients shall be held in one table in one database. Fields that apply to one sex only shall be held in separate tables joined on `patient_id` .

SRS-DBA-06 [M, T, AIMS] Which patients may use a sex-specific table, and which of its fields are mandatory, shall be enforced by database constraints, so a wrong or incomplete record is refused by the database rather than by convention.

Current and previous prescription. These are provided as **views** keyed by `patient_id` , not as two tables. Two real tables keyed by `patient_id` can hold only one previous prescription, so on a patient's third visit the first would be overwritten and lost for good. With views, every visit is kept: the third becomes current, the second becomes previous, and the first is still there.

SRS-DBA-07 [M, I, AIMS] Prescriptions shall be stored in one table linked to their visit, keeping every visit and every version.

SRS-DBA-08 [M, T, AIMS] `current_prescription` and `previous_prescription` shall be provided as views keyed by `patient_id` , resolved by visit date.

SRS-DBA-09 [M, T, AIMS] `previous_prescription` shall return the most recent visit **strictly before** the current one, so a second visit on the same day can never return the visit in progress.

SRS-DBA-10 [M, I, AIMS] Prescription contents shall be stored in typed columns, with one row per medicine in `prescription_items` — drug, dose, frequency and duration each in its own column. Free text stays free text; structured data does not become text.

SRS-DBA-11 [S, I, AIMS] The original JSON shall be kept alongside, in a `JSONB` column, so a field not yet modelled is never lost and can be extracted later.

8.7.3 Keeping both databases safe

SRS-DBA-12 [M, I, AIMS] Table changes shall be made only by versioned scripts held in version control. No one shall change a table by hand.

SRS-DBA-13 [M, T, AIMS] Rules shall be enforced in the database — foreign keys, NOT NULL , CHECK , uniqueness — and not only in application code.

SRS-DBA-14 [M, T, AIMS] Every table holding clinical or identifying data shall have created_at and updated_at , maintained by the database.

SRS-DBA-15 [M, T, AIMS] Access to the clinical database shall be by role, with separate read, write and administration roles. No application shall connect as the owner.

SRS-DBA-16 [M, D, AIMS] Both databases shall have continuous write-ahead-log archiving, and a nightly encrypted dump written to Cloudflare R2, off the UIU server. Continuous archiving protects against a mistake; the off-site dump protects against losing the server.

SRS-DBA-17 [M, D, AIMS] Restoring from backup shall be tested every month, and each test recorded. A backup that has never been restored is not yet a backup.

SRS-DBA-18 [M, T, AIMS] Every read and write of clinical data shall be recorded in an append-only log: who, what and when.

SRS-DBA-19 [S, I, AIMS] Connection pooling shall run in transaction mode, with the database driver configured to match.

8.8 The JSON file beside each recording

One JSON file per visit, with the same name as the WAV, written when the recording is archived and rewritten if the prescription arrives later:

```
{
  "file_stem": "P0012345_DR0042_HOSP003_101432_102847_20260913",
  "patient": { "patient_id": "P0012345", "sex": "female", "full_name": "..." },
  "visit": { "doctor_id": "DR0042", "hospital_id": "HOSP003",
    "cmed_hospital_id": "CMED-DHK-BANANI-01",
    "consent_obtained": true, "consent_method": "verbal_at_reception",
    "start_time": "2026-09-13T10:14:32+06:00",
    "end_time": "2026-09-13T10:28:47+06:00" },
  "paramedic": { "blood_pressure": "120/80", "pulse_bpm": 78 },
  "previous_visit": { "date": "2026-06-02", "prescription": { "...": "..." } },
  "prescription": { "version": 1, "diagnoses": [...], "items": [ { "drug": "..." } ] },
  "recording": { "session_id": "01JB8XQ4M7YZ...", "confirmation": "confirmed",
    "audio_sha256": "..." }
}
```

The database is the index; the JSON file is the portable copy that travels with the audio, including into the cloud copy.

8.9 The operational dashboard — DSH

SRS-DSH-01 [M, D, AIMS] The dashboard shall show daily volume by clinic and by doctor — number of recordings and total recorded hours — as a chart and as a table.

SRS-DSH-02 [M, D, AIMS] It shall let an operator go from clinic, to doctor, to date, to patient, and find a recording at any level.

SRS-DSH-03 [M, D, AIMS] It shall show integrity results: recordings whose chain failed, recordings quarantined and for which of the five causes (§7.5a.1), and recordings stopped by the doctor before the consultation ended.

SRS-DSH-04 [M, D, AIMS] It shall show recording health per room: audio stuck on PCs (SRS-REC-08), speech levels (SRS-LVL-02), and recorders not heard from in three heartbeat intervals.

SRS-DSH-05 [M, D, AIMS] It shall show reconciliation: recordings without a clinical record, and records without a recording (SRS-CRI-10).

SRS-DSH-06 [M, I, AIMS] The dashboard shall read the recordings database only, and shall never show patient names or clinical content.

SRS-DSH-07 [S, D, AIMS] Every count shall open into the rows behind it. A number nobody can look inside is a number nobody can act on.

SRS-DSH-08 [M, D, AIMS] It shall show confirmation results per day and clinic: confirmed, still confirming, and unconfirmed recordings.

SRS-DSH-09 [M, D, AIMS] It shall show backup status: recordings not yet copied to the cloud, pieces in R2 older than 48 hours, and the date and result of the last restore test.

SRS-DSH-03 changes behaviour, not just reporting. Showing quarantines and forced stops against clinic and doctor turns a recurring problem in one room into a visible pattern, instead of a series of unrelated incidents.

9. Non-functional requirements

9.1 Speed

ID	Requirement	Pri	Ver
SRS-NFP-01	Trigger to microphone open: 500 ms or less, for 95% of consultations.	M	A
SRS-NFP-02	Trigger to the reply 200 RECORDING_STARTED : 2 s or less, for 95%, on a working connection.	M	A
SRS-NFP-03	Handover between patients: no gap in capture, measured with a continuous test tone.	M	T
SRS-NFP-04	Recorder processor use: under 5% of one core, on average, while recording.	S	A
SRS-NFP-05	Recorder memory: under 250 MB.	S	A
SRS-NFP-06	AIMS LAB server reply time for control requests: under 300 ms for 95%, not counting piece checks.	S	A
SRS-NFP-07	Piece check on commit, including read-back and fingerprint: under 3 s for 95%.	S	A

9.2 Volume

ID	Requirement	Pri	Ver
SRS-NFC-01	7 clinics, 14 consulting rooms, 30 doctors across two shifts, about 560 sessions and 112 recording hours a day. At most 14 recordings run at once — one per room.	M	A
SRS-NFC-02	About 36 GB of audio a day to the archive (11 TB a year), and about 16 GB a day of FLAC copies to the cloud (5 TB a year, planning figure).	M	A
SRS-NFC-03	Absorb three times the normal load for one hour without loss, using the buffer.	M	A
SRS-NFC-04	Support twice today's size without changing the design.	S	A

9.3 Availability and reliability

ID	Requirement	Pri	Ver
SRS-NFR-01	The recorder shall keep recording through a short network interruption and deliver the backlog automatically when the connection returns.	M	D
SRS-NFR-02	UIU server availability target: 99.5% a month, excluding planned maintenance, which shall be scheduled outside clinic hours.	S	A
SRS-NFR-03	Updating the AIMS LAB server shall not interrupt any recording in progress.	M	D
SRS-NFR-04	No single failure shall lose audio silently; every way of losing audio shall raise an alert.	M	T
SRS-NFR-05	Archived audio shall never be lost: the archive is on RAID 6 and every recording has a verified cloud copy.	M	A
SRS-NFR-06	If the UIU server is lost entirely, the archive shall be rebuildable from the cloud copies, and the databases from their nightly dump.	M	D

9.4 Security

ID	Requirement	Pri	Ver
SRS-NFS-01	Protection shall be layered: page address, Host check, loopback, grant, confirmation against API 2, device token, hash chain, read-back check, receipt, Channel B key. No single layer's failure shall be enough to cause harm.	M	I
SRS-NFS-02	Credentials shall be compared in constant time.	M	I
SRS-NFS-03	AIMScribe_Agent.exe and AIMScribeSetup.exe shall be code-signed (Authenticode).	M	I
SRS-NFS-04	Every credential shall be replaceable without reinstalling recorders, except the device key, which needs re-enrolment by design.	S	D
SRS-NFS-05	Security failures shall be logged with enough detail to diagnose, and without revealing secrets.	M	I
SRS-NFS-06	The system shall treat the browser page as hostile, and take no trust from its claims alone.	M	I

9.5 Monitoring

ID	Requirement	Pri	Ver
SRS-NFO-01	Every recorder shall send a heartbeat every 30 s, with its version, buffer depth and number of waiting pieces.	M	T
SRS-NFO-02	A recorder not heard from for three heartbeat intervals during clinic hours shall raise an alert.	M	T
SRS-NFO-03	Alerts shall exist for: chain failure, fingerprint mismatch on arrival, buffer above 25%, abnormal close, silent session, repeated gate refusal, unconfirmed recordings, Channel B errors, archive or cloud-copy backlog, pieces in R2 older than 48 hours, disk space on the UIU server, and certificate expiry.	M	T
SRS-NFO-04	Operators shall be able to see the state of every clinic without database access.	S	D
SRS-NFO-05	Logs shall be structured and shall contain no patient information beyond the pseudonymous <code>patient_id</code> .	S	I

9.6 Maintainability

ID	Requirement	Pri	Ver
SRS-NFM-01	The message format version shall be explicit; a mismatch shall be refused, never guessed at.	M	T
SRS-NFM-02	Fixed reference messages with pinned fingerprints shall exist in both repositories, to catch an incompatible change.	M	T
SRS-NFM-03	The AIMS LAB server shall be deployable from the same container images on the UIU server or on a cloud host.	M	D
SRS-NFM-04	No part shall depend on one provider's special features without a layer that allows the provider to be replaced.	S	I
SRS-NFM-05	Unknown fields in a message shall be ignored or stored, never rejected, so either side can add fields without breaking the other.	M	T

9.7 Ease of use

ID	Requirement	Pri	Ver
SRS-NFU-01	A doctor shall need no training beyond "the red circle stops it, the blue rectangle pauses it".	M	D
SRS-NFU-02	Installation shall be one signed installer, three fields, two of them already filled.	M	D
SRS-NFU-03	The tray icon shall show the recording state at a glance.	S	D
SRS-NFU-04	Every message a doctor sees shall be plain language, with no technical words.	M	I

10. Capacity — the UIU server and cloud storage

10.1 The load

Figure	Value	Source
Clinics	7	6 Aalo branches and Amader Susastho
Consulting rooms	14	Deployment plan
Recordings at the same time, at most	14	One per room
Enrolled laptops	16	One per room, plus two spares
Doctors	30	Two shifts share the same rooms
Room hours per day	8	Morning and afternoon shifts
Recording hours per day	112	14 rooms × 8 hours
Clinic days per year	312	Six days a week
Average consultation	12 minutes	Estimate — OD-05
Sessions per day	About 560	Derived
Audio	318 MB per recorded hour	WAV, 44.1 kHz, mono, 16-bit (SRS-CAP-01)

10.2 What that means

Quantity	Value
Pieces per day	About 9,000
Recorder requests per day	About 28,000
Channel B requests per day	About 1,100 — API 2 and a prescription for each visit
Average request rate	About 1.2 per second
Peak request rate	About 4 per second
Audio uploaded to R2	About 36 GB a day
Read back by the UIU server at the busiest time	About 10 Mbit/s
Archive growth	About 36 GB a day, 11 TB a year
Cloud copy growth, planning figure	About 16 GB a day, 5 TB a year
Cloud copy upload	About 16 GB each night — under an hour at 50 Mbit/s

This is not a processor-heavy system. It is a storage and bandwidth system.

About one request a second is trivial for any server. 36 GB of audio every clinic day is not trivial, and it arrives whether anyone is watching or not.

Size by rooms, not by doctors. A room records one consultation at a time, so 30 doctors in 14 rooms produce 14 simultaneous recordings, not 30. Every figure above follows from the room count.

10.3 The UIU server

Figure 2 in §3.2 shows what runs on it.

Part	Minimum	Recommended
Processor	8 cores	16 cores , server grade, with AES-NI and SHA extensions
Memory	32 GB ECC	64 GB ECC
System disks	2 × 480 GB SSD, mirrored	2 × 960 GB NVMe, mirrored
Database disks	2 × 1 TB NVMe, mirrored, power-loss protected	2 × 2 TB NVMe, mirrored, power-loss protected
Working disks	1 × 1 TB NVMe	2 × 2 TB NVMe, mirrored
Archive array	6 × 12 TB, RAID 6 — 48 TB usable, about 3.5 years	8 × 16 TB, RAID 6 — 96 TB usable, about 7 years
Internet	50 Mbit/s up and down	100 Mbit/s up and down
Power	UPS, 30 minutes	UPS, 30 minutes, with automatic safe shutdown
Software	Ubuntu Server 24.04 LTS, Docker Compose, Caddy or Nginx, PostgreSQL 16, PgBouncer	Same

Archive years assume 11 TB a year with 20% of the array kept free. Recording at 48 kHz (**OD-12**) adds 9% to every storage figure.

Why each part matters.

- **ECC memory** corrects memory errors by itself. For a system whose whole purpose is proving recordings were not changed, it is required.
- **Power-loss-protected database disks** decide whether the system feels fast and whether its records are true. Cheap SSDs report a write as saved while it is still in volatile memory, and lose it in a power cut — the exact failure the hash chain exists to detect.
- **Separate working disks** hold pieces downloaded for checking and for merging, so a busy merge never slows a database query a recorder is waiting on.
- **RAID 6** keeps working if any two disks fail at once. With large disks, a second failure during a rebuild is common enough that RAID 5 is not safe.
- **A symmetric internet line** is needed because the server downloads every piece during clinic hours and uploads the cloud copy at night. Most business lines are fast down and slow up.
- **The UPS must shut the server down cleanly**, or a recording being written when the power fails can be damaged.

What a user notices	Target
Doctor opens a patient → microphone on	Under 0.5 s — happens on the PC
Recorder asks permission	Under 0.3 s
One piece accepted and checked	Under 1 s
Dashboard page loads	Under 1 s
Finished consultation appears in the archive	Within 5 minutes
Cloud copy made	The same night

SRS-SRV-01 [M, I, AIMS] The UIU server shall use ECC memory.

SRS-SRV-02 [M, A, AIMS] Database disks shall be mirrored NVMe with power-loss protection, with a measured flush time under 1 ms for 99% of writes.

SRS-SRV-03 [M, I, AIMS] Working disks shall be separate from the database disks and from the archive array.

SRS-SRV-04 [M, D, AIMS] The archive array shall keep working with any two disks failed, shall be encrypted at rest, and shall be readable only by the AIMSScribe services and named staff.

SRS-SRV-05 [M, A, AIMS] The server's internet line shall provide at least 50 Mbit/s in both directions (**OD-18**).

SRS-SRV-06 [M, D, AIMS] The server shall be on a UPS that shuts it down cleanly before the battery runs out.

SRS-SRV-07 [M, I, AIMS] From the internet, only port 443 shall be open. Remote administration (SSH) shall be allowed only from named machines on the UIU network.

SRS-SRV-08 [M, D, AIMS] Monitoring shall alert on disk space, failed background jobs, silent recorders and certificate expiry, before a doctor would notice.

SRS-SRV-09 [M, I, AIMS] Server configuration shall be kept in version control, so a replacement server can be built from it.

10.4 Cloud storage

Lossless or lossy — measured on our own recordings. 48 real recordings, spread across 12 recording days, 7.2 hours of audio:

	FLAC (lossless)	Opus 32 kbps (lossy)
Size of our recordings	16% of the WAV overall — 18% typical, 28% or less for 9 in 10, 42% at most	About 4%
Decodes back to the original exactly	Yes — every file checked, every sample identical	No
Can be checked against the hash chain	Yes	No
Can rebuild the UIU archive	Yes	No
Speed	About 1,100 times faster than real time — a day's audio in about six minutes of one core	—

Why we plan with 45%, not 16%. Some of today's recordings contain stretches of digital silence caused by the speakerphone's noise gate (§7.1a). Silence compresses extremely well, which flatters today's figure. Once microphones are fixed, recordings will contain real room sound and compress less. 45% is a cautious figure for clean speech.

	Per recorded hour	Per day	Per year
Original WAV (on the UIU archive)	318 MB	36 GB	11.1 TB
FLAC, planning figure (45%)	143 MB	16 GB	5.0 TB
FLAC, 9 in 10 of today's files (28%)	89 MB	10 GB	3.1 TB
Pieces waiting in R2 to be merged	—	36 GB, deleted within about a day	About 70 GB at any moment

Cost. Monthly bill at the end of each year, at 5 TB a year. Published list prices, checked on 13 September 2026. Nothing needs to be bought in advance; every provider charges monthly for what is stored.

Provider	Per TB per month	End of year 1	End of year 2	End of year 3	About 3 years in total
Cloudflare R2, Standard	\$15	\$75	\$150	\$225	\$4,000
Cloudflare R2, Infrequent Access	\$10	\$50	\$100	\$150	\$2,700
Backblaze B2	\$6.95	\$35	\$70	\$104	\$1,900
Wasabi	\$7.99	\$40	\$80	\$120	\$2,200
Amazon S3, Standard	\$23	\$115	\$230	\$345	\$6,200
Amazon S3, Glacier Deep Archive	\$0.99	\$5	\$10	\$15	\$270, plus retrieval charges

Upload and request charges add about \$2 a month on R2. Downloading a year of copies (5 TB) costs nothing on R2, and about \$450 on Amazon S3 — which matters because transcription and research read audio back. Glacier Deep Archive looks cheapest but takes hours to retrieve, charges per GB, and keeps every file at least 180 days; it does not suit audio the pipeline will read.

Choice: Cloudflare R2 Standard. The system already works with it, downloads are free, and billing is simple. Revisit at 10 TB, around the end of year 2. If cost becomes the priority, files older than 30 days can move to R2 Infrequent Access, or the copies can move to Backblaze B2 (SRS-ST0-05).

Safety comes from settings, not the provider. All of these providers state the same durability and encrypt at rest. What protects patients is: encrypting before upload with a key that stays at UIU (SRS-DAT-08); locking the copy bucket (SRS-ARC-12); and giving each role its own limited credential (SRS-ST0-02). None of these providers stores data in Bangladesh — **OD-14**.

10.5 Upload needed at each clinic

Rooms at the site	Upload needed
2 (every site today)	About 1.4 Mbit/s
3	About 2.1 Mbit/s
4	About 2.8 Mbit/s

Modest, but sustained through clinic hours. The buffer covers short gaps; the line does not need to be sized for the worst moment.

10.6 Growth

Scale	Rooms	Audio per day	Archive per year	Cloud copy per year	What changes
Today	14	36 GB	11 TB	5 TB	—
2×	28	71 GB	22 TB	10 TB	Storage only
5×	70	178 GB	55 TB	25 TB	Storage, a larger internet line, and a second API instance
10×	140	356 GB	111 TB	50 TB	Tiered storage; regional servers

At every scale, the first limit is how long audio is kept, not processing power (SRS-DAT-07 , **OD-06**).

11. Failure modes

#	Failure	How it is noticed	What happens	Doctor sees	Effect on care
1	Recorder not installed	Connection refused	CMED page logs it and carries on	Nothing	None
2	Recorder stopped	Connection refused	Page reconnects with growing delay	Nothing	None
3	UIU server unreachable from the clinic	Requests time out	Keeps recording; pieces wait in the buffer (4 GB, about 13 hours) and upload when it returns	Tray warning only	None
4	Clinic internet down	Same as 3	Same as 3	Tray warning only	None
5	Cloudflare R2 unreachable	Uploads fail	Same as 3	Tray warning only	None
6	Buffer over 25%, or a piece waiting over 15 minutes	Checked on the PC	Alert; recording continues	Tray warning	None yet
7	Buffer full	Checked on the PC	New sessions refused until space frees	Clear message	Recording lost; consultation unaffected
8	Microphone lost mid-session	Capture error	Warn, try to recover, keep what exists	Warning on the control	Partial recording
9	PC not enrolled or revoked, clinic mismatch, or doctor not at clinic	Server reply	Refused; captured audio discarded (<code>SRS-GRT-08</code>); alert	Brief message	Recording lost — deliberately
10	PC clock wrong by more than 65 s	Grant check fails	Refused	"Authorisation failed"	Recording lost until the clock is fixed
11	API 2 late	No matching notice yet	Keeps recording; asks every 5 s	"Confirming"	None
12	API 2 never arrives	Two minutes without a match	Keeps recording; marked unconfirmed; alert. Admitted if API 2 arrives within 24 h, otherwise deleted.	Nothing more	Recording withheld from the dataset
13	CMED server cannot reach the UIU server	CMED sees timeouts	CMED retries in the background (<code>SRS-CHB-05</code>)	Nothing	Confirmation delayed
14	Fingerprint mismatch on arrival	Read-back check	Re-upload to a new object; quarantine if it persists; never lost (§7.5a)	Nothing	Resolved automatically

#	Failure	How it is noticed	What happens	Doctor sees	Effect on care
15	Chain fails at close	Server check	Quarantine and alert	Nothing	Resolved automatically
16	Power cut mid-piece	Journal on restart	Half-written piece rejected; at most 99 s lost	Nothing	Small, bounded loss
17	Prescription never built	Gate stays unarmed	Session stays open; Stop always available	Control stays visible	Long session, reported
18	Two triggers in quick succession	Gate state	Second refused while unarmed	Nothing	None
19	Archive or cloud-copy worker stopped	Backlog alert	Pieces stay in R2; nothing deleted early	Nothing	None — delayed
20	A cloud-copy step fails	Step check	Pieces kept; step retried; alert if it repeats (SRS-ARC-10)	Nothing	None
21	One or two archive disks fail	RAID alert	Array keeps working; disks replaced	Nothing	None
22	UIU server lost entirely	Monitoring	Archive rebuilt from cloud copies, databases from the nightly dump (SRS-NFR-06); recorders buffer meanwhile	Tray warning if prolonged	None for archived audio; buffered audio at risk after about 13 hours
23	Patient did not consent	API 2 has <code>consent_obtained: false</code> , or no consent field	Recording stopped; captured audio deleted without upload; no clinical content kept (SRS-CNF-11 , SRS-CNF-12)	"This consultation is not being recorded"	None — as intended

Read the last column. In these twenty-three situations, a recording may be delayed, withheld, lost or deliberately not kept — but in none of them is a doctor stopped from seeing a patient. That is SRS-IF1-17 holding all the way down.

Part IV — Closing out

12. Verification and acceptance

12.1 Acceptance tests

Each test passes or fails on a running system, and names the requirements it checks.

ID	Test	Checks	Passes when
AT-01	Doctor opens a patient in CMED	IF1-01, CAP-03, GRT-07	Microphone on within 500 ms; 200 RECORDING_STARTED within 2 s
AT-02	Send a trigger with no doctor_id	INV-03	Refused 400 MISSING_FIELD ; nothing recorded
AT-03	Send API 2 with consent_obtained: false , then the trigger	BKD-01, CNF-11, CNF-12, UIX-14	Recording stops; no audio uploaded; no clinical content stored; both Channel B replies 200 CONSENT_NOT_GIVEN
AT-04	Send a trigger naming a clinic other than the PC's	GRT-10, ENR-20	Refused 401 CLINIC_MISMATCH ; alert raised
AT-05	Connect from a page address not on the allowlist	IF1-06	Connection closed with 4403 before it is accepted
AT-06	Connect with Host: evil.example pointing at 127.0.0.1	IF1-06	Refused
AT-07	Replay a copied grant within 60 s	GRT-05	Refused
AT-08	Present a grant 90 s old	GRT-04	Refused as expired
AT-09	Trigger patient B while recording patient A, gate unarmed	GAT-03	409 GATE_NOT_ARMED ; patient A's recording continues
AT-10	Send prescription_built , then trigger patient B	GAT-04, SES-02, SES-03	A closes, B opens; a continuous test tone shows no gap
AT-11	Send prescription_built naming the wrong patient or session	IF1-13	409 PATIENT_MISMATCH ; gate not armed
AT-12	Send prescription_built twice	IF1-14	Second reply 200 GATE_ALREADY_ARMED
AT-13	Press Stop, then try to close the reason form	UIX-06, UIX-07	Form cannot be closed; session stays open
AT-14	Press Stop and give a reason	UIX-06, UIX-08	Microphone cuts on the press; session closes on confirm; reason in the chain
AT-15	Press Pause without a reason	UIX-05	Pause does not take effect
AT-16	Disconnect the clinic's internet for 4 hours of recording	NFR-01, UPL-06	All audio buffered; uploads complete when the link returns; nothing lost
AT-17	Cut power mid-piece	SPL-04, SPL-05	Half-written piece rejected on restart; journal consistent; at most 99 s lost
AT-18	Corrupt a piece in R2 before commit	UPL-02, UPL-03	Server detects it on read-back
AT-19	Remove one chain entry before close	CHN-05	Chain fails; session quarantined
AT-20	Fill the buffer past 25%, then to its 4 GB limit	SPL-06, SPL-07	Alert at 25%; new sessions refused when full

ID	Test	Checks	Passes when
AT-21	Redeem an enrolment code twice from the same never-seen device	ENR-08	Second attempt succeeds with re-issued credentials
AT-22	Redeem a code twice after the device has sent a heartbeat	ENR-08	Refused; replay logged
AT-23	Redeem an expired code	ENR-03, ENR-04	Refused with the same message as unknown and used codes
AT-24	Enrol while the server is unreachable	ENR-07	Code stays on disk; later retry succeeds
AT-25	Wipe the key folder and restart	ENR-11	Recorder refuses to record and asks for re-enrolment
AT-26	Revoke a device, then try to upload	ENR-15, ENR-16	Refused at every route
AT-27	Stop the recorder, then open a patient in CMED	IF1-17	CMED logs it and carries on; no dialog; doctor unaffected
AT-28	Set the PC clock 10 minutes fast	ASM-07, GRT-04	Recording refused; log names clock difference as the cause
AT-29	Run 14 simulated rooms for one clinic day	NFC-01, NFP-06	No lost pieces; reply times within §9.1
AT-30	Update the AIMS LAB server during a recording	NFR-03	No interruption; no lost piece
AT-31	Send a 128 KB message on Channel A	IF1 connection	Refused; connection kept
AT-32	Send a message with an unknown extra field	NFM-05	Ignored; command succeeds
AT-33	Full path: consultation → archive → cloud copy	ARC-01–04, ARC-09, SES-05, REC-16	Correct file name; chain verified; receipt issued and PC copy deleted at once; WAV and JSON archived; FLAC copy stored; pieces deleted
AT-34	Record 60 s of an occupied room and count digital silence	CAP-11	Under 0.5%; otherwise the room is not commissioned
AT-35	Compare capture through each Windows audio path in the same room	CAP-07, CAP-08	The native low-latency path is chosen and is the least gated
AT-36	Speak loudly from the doctor's chair and quietly from the patient's	LVL-01, LVL-03	Quiet-speaker prompt appears; loud-speaker prompt does not
AT-37	Mute the microphone at its switch mid-session	LVL-04, CAP-06	Reports a microphone problem, not a patient problem
AT-38	Record a clinic day in a commissioned room and measure piece lengths	SEG-07	At least 80% are 30–60 s
AT-39	Feed a recording whose speech is below the fixed quiet threshold	SEG-06	Pieces still fall in the target window

ID	Test	Checks	Passes when
AT-40	Re-enrol a device with pieces still in its buffer	ENR-21	Refused until the buffer is empty
AT-41	Compare the archived WAV byte for byte with the captured audio	DAT-11	Identical; SHA-256 matches the chain
AT-42	Corrupt a piece in transit, leaving the local file intact	REC-01, REC-02	Local check passes; re-uploaded to a new object; session completes
AT-43	Corrupt the local piece on disk, then let it upload	REC-01, REC-04	Retries stop; alert names local damage; piece delivered to quarantine
AT-44	Use up the retry allowance on one piece	REC-03, REC-05	Exactly three attempts; session quarantined only after the third
AT-45	Clear a quarantine after a verified re-upload	REC-06	Session archives normally
AT-46	Leave a session quarantined and read the heartbeat	REC-08, REC-09	Stuck audio reported centrally and shown in the operator view
AT-47	Send API 2 with demographics and a previous prescription	CRI-01, CRI-02	Both stored; recording unaffected
AT-48	Withhold API 2 and the prescription for one consultation	CRI-04, CRI-10, CNF-08	Recording completes, unconfirmed; the nightly check reports the gap
AT-49	Send a record that fails validation	CRI-07, CHB-10	422 SCHEMA_INVALID ; stored unchanged in quarantine; nothing partly loaded
AT-50	Send the same record twice	CRI-09, CHB-07	One row; second reply 200 ALREADY_RECEIVED
AT-51	Search archive paths, file names, object names, logs and the chain for the patient's name	CRI-08, DAT-02, INV-04	Not found in any of them
AT-52	Record a third visit for one patient	DBA-07, DBA-08	All three kept; views return the right current and previous
AT-53	Two visits for one patient on the same day	DBA-09	previous_prescription returns the earlier one, never the visit in progress
AT-54	Save female-only details for a male patient	DBA-06	Refused by a database constraint
AT-55	Find recordings for one clinic, doctor and date	DBA-02, DSH-02	Answered from the index, with no folder search
AT-56	Open the dashboard as an operations user	DSH-06	Volumes, integrity and reconciliation visible; no patient name or clinical content reachable
AT-57	Send the trigger and a matching API 2 together	CNF-02, CNF-04	Recording confirmed and linked to its patient information
AT-58	Trigger the recorder with no API 2 sent	CNF-04, CNF-08	Recording continues; unconfirmed after two minutes; alert; kept out of the dataset
AT-59	Send API 2 thirty seconds after the trigger	CNF-06, CNF-07, UIX-13	Microphone on from the start; control shows "confirming"; confirmed when API 2 arrives

ID	Test	Checks	Passes when
AT-60	Send API 2 an hour after a recording was marked unconfirmed	CNF-09	Recording admitted
AT-61	Send API 2 and never trigger	CNF-03	Notice expires after five minutes; no error
AT-62	Send API 2 with <code>start_time</code> one second different from the trigger	CNF-02	No match — both must use the single server-created value
AT-63	Send API 2 for a hospital other than the PC's clinic	CNF-04	No match
AT-64	Archive one consultation end to end	ARC-08, ARC-09	WAV and JSON at UIU; FLAC copy in the locked bucket; pieces deleted only after both are verified
AT-65	Stop the worker between the upload and the database record	ARC-09, ARC-10	Pieces still present; the step is retried and completes on restart
AT-66	Corrupt the FLAC before its decode check	ARC-09, ARC-10	Mismatch detected; pieces kept; alert raised
AT-67	Try to delete a copy using the piece clean-up credential	ARC-12, STO-02	Refused
AT-68	Call Channel B with a missing or wrong key	CHB-02	<code>401 INVALID_KEY</code> ; nothing stored
AT-69	Send a prescription, then a changed one for the same five fields	CHB-08, ARC-13	Both versions kept; latest is current; JSON rewritten and re-copied
AT-70	Send the prescription before the patient information	CHB-09	Accepted, and attached when API 2 arrives
AT-71	Make the server return <code>503</code> for ten minutes while CMED sends API 2	CHB-05, CHB-06, IF1-17	CMED retries in the background; doctor unaffected; recording confirmed on delivery
AT-72	Search the CMED page, its network traffic and CMED's repository for the API key	CHB-01	Not found
AT-73	Stop the UIU server for one hour during clinic	NFR-01, SPL-06	Recording continues; buffers drain when it returns; nothing lost
AT-74	Restore one day's recordings from the copy bucket to a spare machine	STO-06, NFR-06	Decoded audio identical to the archive; chains verify
AT-75	Leave a recording unconfirmed for 24 hours	CNF-09, ARC-16	Never archived or copied; pieces deleted from R2; deletion in the audit log
AT-76	Fail two disks in a test RAID 6 array during archiving	SRV-04	Archiving continues; alert raised
AT-77	Send API 2 with no consent field	CNF-11	Treated exactly as no consent

12.2 Pilot exit criteria

The integration is accepted when:

1. Every test in §12.1 passes on a clinic-representative PC and the UIU server, except any tied to an open decision that has been deferred in writing.
2. Twenty consecutive real consultations at one clinic are confirmed, archived with verified chains and correct file names, and copied to the cloud.
3. There are zero incidents in which AIMSCRIBE stopped a doctor from working.
4. One power-cut test, one internet-outage test and one restore test pass in the field, not only on the bench.
5. Every **OD-~~nn~~** is closed, or deferred in writing.

12.3 Which tests check which requirements

Requirement group	Checked by
INV	AT-02, AT-04, AT-51, code inspection
ENR	AT-21 – AT-26, AT-40
GRT	AT-04, AT-07, AT-08, AT-28
CNF	AT-03, AT-57 – AT-63, AT-75, AT-77
IF1	AT-01, AT-05, AT-06, AT-09 – AT-12, AT-27, AT-31, AT-32
CHB	AT-49, AT-50, AT-68 – AT-72
STO	AT-67, AT-74
CAP · LVL · SEG	AT-34 – AT-39, commissioning measurement
SPL · UPL · CHN	AT-16 – AT-20, AT-73
REC	AT-33, AT-42 – AT-46
SES · GAT · UIX	AT-09 – AT-15, AT-59
BKD · ARC	AT-03, AT-30, AT-33, AT-64 – AT-67, AT-69, AT-75
DAT · CRI	AT-41, AT-47 – AT-51
DBA · DSH	AT-52 – AT-56
SRV · NF*	AT-29, AT-30, AT-73, AT-74, AT-76, measurement against §9 and §10

13. Delivery plan and division of work

13.1 What works today, and what is still to build

This document is a design baseline, not a description of finished software.

Working today, tested with the dummy application	Owner
Channel A connection with page-address, <code>Host</code> and loopback checks	AIMS — done
Enrolment, device tokens, revocation, audit log	AIMS — done
Recording, cutting into pieces, encrypted buffer, hash chain	AIMS — done
Upload to R2 through upload links, and read-back checking	AIMS — done
Seamless handover between patients	AIMS — done
Grant checking in the recorder: Ed25519, 60 s, single use	AIMS — done

Still to build	Owner
Five-field trigger, <code>prescription_built</code> , and the gate (§6.1, §7.7)	AIMS
Replies carrying <code>request_id</code> , <code>status</code> and <code>code</code> (§6.1.4)	AIMS
POST <code>/api/v2/grant/mint</code> on the server (§5)	AIMS
Channel B endpoints: <code>patient-information</code> and <code>prescription</code> (§6.2)	AIMS
Confirmation matching (§5.6)	AIMS
The on-screen Stop and Pause control (§7.8)	AIMS
Receipts on custody, and clean-up at start-up and shutdown (<code>SRS-REC-15</code> — <code>18</code>)	AIMS
Both databases and the JSON beside each recording (§8.7, §8.8)	AIMS
The cloud-copy worker and the two buckets (§7.10, §6.4)	AIMS
The UIU server: purchase, installation, and moving the backend from Render and Neon (§10.3)	AIMS, with UIU IT
The dashboard (§8.9)	AIMS
A test page and a test Channel B environment for CMED (<code>SRS-API-04</code> , <code>SRS-API-05</code>)	AIMS
The intermediate save point in the buffer (<code>SRS-SPL-10</code> — <code>12</code>)	AIMS
API 1, API 2 and API 3 in CMED's page and server	CMED

AIMS LAB builds its side first. The recorder changes, the Channel B endpoints and the test page are finished and running before CMED starts (Phase 1 below), so CMED never builds against something that does not exist yet.

13.2 CMED's work, item by item

#	Task	Estimate
1	Give AIMS LAB the exact page addresses, production and testing	10 minutes
2	Give AIMS LAB the clinic identifiers, and which AIMS LAB clinic each maps to	30 minutes, once
3	Confirm that <code>patient_id</code> never changes for a patient	—
4	Open the Channel A connection from the page and keep it open (Appendix B.1)	Half a day
5	Send API 1 when patient details open, using <code>start_time</code> from the server	Half a day
6	Send <code>prescription_built</code> when the prescription is built	Quarter of a day
7	Send API 2 from the server at the moment the patient is opened, with the consent from reception records, and the prescription when it is built, with background retry (Appendix B.3)	One day
8	Handle replies by <code>code</code> , and ignore failures quietly	Half a day
9	Joint testing with AIMS LAB	One day
	Total	3–4 developer-days

13.3 What AIMS LAB brings

Working recorders on enrolled PCs, the UIU server, both databases, the archive and the cloud copy, the ASR pipeline downstream, installers and support, every key and its rotation, the Channel B endpoints and test environment, server and storage capacity and its cost, and this specification.

13.4 Order of work

Phase	What	Moves on when
0	Agree page addresses, clinic mapping, identifier stability, the consent wording (<code>SRS-DAT-15</code>), and the open decisions; exchange the Channel B test key	Written agreement
1	AIMS LAB: recorder changes, grant, Channel B endpoints, confirmation, test page and test environment	<code>AT-01</code> – <code>AT-12</code> , <code>AT-57</code> – <code>AT-63</code> and <code>AT-68</code> – <code>AT-70</code> pass on a bench
2	CMED: tasks 4–8 against the test environment, following the testing steps in the <i>Integration Guide</i>	<code>AT-27</code> , <code>AT-32</code> , <code>AT-71</code> , <code>AT-72</code> pass
3	AIMS LAB: on-screen control; UIU server in service; archive and cloud copy	<code>AT-13</code> – <code>AT-15</code> , <code>AT-64</code> – <code>AT-67</code> , <code>AT-73</code> , <code>AT-74</code> pass
4	Together: one clinic, one week	§12.2 exit criteria
5	Roll out clinic by clinic	Sign-off per clinic

14. Open decisions

ID	Decision	Needed from	Blocks	If it is late
OD-01	Exact production and testing page addresses for the allowlist	CMED	Phase 2	Nothing can connect
OD-02	Is <code>patient_id</code> guaranteed never to change for a patient?	CMED	Phase 0	One patient's recordings split into unrelated sets
OD-03	Clinic mismatch: refuse (recommended), or record with a warning during the pilot?	Joint	Phase 1	Default is refuse (<code>SRS-GRT-10</code>)
OD-04	What share of consultations end without a prescription being built?	CMED and clinical team	Phase 3	Decides whether the gate also needs a time limit
OD-05	Average consultation length — measured, not estimated	Clinical team	Phase 4	§10 sizing scales with it
OD-06	How long audio is kept, on the archive and in the cloud	AIMS LAB and ethics committee	Buying storage	11 TB a year on the archive and 5 TB in the cloud, growing every year
OD-07	The list of Stop and Pause reasons, and which count as abnormal	Clinical team	Phase 3	The on-screen control cannot be finished
OD-08	Does a long pause need a supervisor's name?	Clinical team	Phase 3	Affects <code>SRS-UIX-10</code>
OD-09	Save-point interval in the buffer	Measurement	Phase 3	Must come from the disk benchmark, not a guess
OD-10	<i>Closed 13 September 2026:</i> the AIMS LAB server is hosted at UIU (§10.3).	—	—	—
OD-11	Can the current speakerphone meet §7.1a, or must the microphone be replaced?	Measurement	Commissioning	Decided by <code>AT-34</code> and <code>AT-35</code> , not by opinion
OD-12	Record at the microphone's native 48 kHz instead of 44.1 kHz?	AIMS LAB	Phase 1	Removes a resampling step; costs 9% more storage
OD-13	<i>Closed 14 September 2026:</i> consent is sent in API 2 as <code>consent_obtained</code> and <code>consent_method</code> , from CMED's reception records (§6.2.2, <code>SRS-CNF-11</code>).	—	—	—
OD-14	Is storing encrypted patient audio with a provider outside Bangladesh acceptable under the study's approval and national data-protection rules?	Ethics committee	First cloud upload	No patient audio can go to the cloud
OD-15	The fields for <code>female_details</code> and <code>male_details</code>	Clinical team	Clinical database	Those tables cannot be finished

ID	Decision	Needed from	Blocks	If it is late
OD-16	The full list of paramedic measurements CMED will send	CMED	Phase 1	API 2 schema incomplete
OD-17	How long patient information is kept	Ethics committee	Clinical database	Retention of clinical records undefined
OD-18	UIU internet line (100 or 50 Mbit/s symmetric), fixed public address and domain name, rack and UPS, and who is on call	UIU IT	Phase 3	The server cannot go into service
OD-19	Who is in the clinical access group	Principal investigator	Clinical database	No one can be given clinical access
OD-20	How long the copy bucket locks each object against deletion	AIMS LAB	First cloud copy	Follows OD-06
OD-21	Minimum or recommended archive array (48 TB or 96 TB)	AIMS LAB and UIU IT	Buying hardware	About 3.5 or 7 years before more disks are needed

15. Appendices

Appendix A — Response codes

A.1 Reply to `start` (API 1)

status	code	Meaning	What CMED should do
200	RECORDING_STARTED	Recording; <code>session_id</code> returned	Keep the <code>session_id</code>
202	RECORDING_PROVISIONAL	Recording; permission still being checked	Keep the <code>session_id</code> ; a later event reports the result
400	MISSING_FIELD	A required field is missing	Fix the message — a build-time bug
400	INVALID_IDENTIFIER	An identifier has characters that are not allowed	Fix the identifier
401	AUTHORISATION_FAILED	Permission could not be granted	Log it; do not retry automatically
401	CLINIC_MISMATCH	This PC does not belong to that hospital	Log it and tell AIMS LAB — the mapping is wrong
403	ORIGIN_NOT_ALLOWED	The page address is not on the allowlist; the connection is closed with <code>4403</code>	Configuration — tell AIMS LAB the address
404	DOCTOR_NOT_AT_CLINIC	The doctor is not registered at this clinic	Log it; AIMS LAB updates the register
409	GATE_NOT_ARMED	A consultation is still open	Expected. Do nothing.
409	SESSION_ALREADY_ACTIVE	This exact session is already recording	Ignore — usually a double click
423	DEVICE_NOT_ENROLLED	This PC is not registered	Tell AIMS LAB which machine
503	AGENT_NOT_READY	The recorder is starting or recovering	Retry once after a moment
—	<i>no connection</i>	Recorder not installed or not running	Log it and carry on. Never block the doctor.

A.2 Reply to `prescription_built` (API 3, part one)

status	code	Meaning	What CMED should do
200	GATE_ARMED	Accepted; the next trigger may hand over	Nothing
200	GATE_ALREADY_ARMED	Already armed	Nothing
400	MISSING_FIELD	<code>patient_id</code> or <code>session_id</code> missing	Fix the message
409	NO_ACTIVE_SESSION	Nothing is recording	Ignore
409	PATIENT_MISMATCH	Names a patient or session other than the open one	Log it — worth investigating

A.3 Replies on Channel B (API 2 and API 3, part two)

See §6.2.4. In short: `202` and `200` need nothing; `400` , `401` , `413` and `422` are logged for a developer and not retried unchanged; `429` and `5xx` are retried in the background.

A.4 Error message on Channel A

```
{
  "event": "error",
  "command": "start",
  "request_id": "cmed-7f3a2b91",
  "status": 409,
  "code": "GATE_NOT_ARMED",
  "message": "The current consultation has not been completed yet.",
  "timestamp": "2026-09-13T10:14:32+06:00"
}
```

Act on `code`, never on `message`. Message text is written for doctors and may be reworded; codes are part of the contract.

Appendix B — Reference code

B.1 The page's connection to the recorder

This is the dummy application's client, reduced to what CMED needs. Every method returns a result instead of throwing, so nothing in it can break the CMED page.

```

// aimscribe.ts – connection from the CMED page to the recorder on this PC.
type Reply = { status: number; code: string; data?: any };

export class AimScribe {
  private ws: WebSocket | null = null;
  private waiting = new Map<string, (r: Reply) => void>();
  private backoff = 1000;

  connect(): void {
    try {
      const ws = new WebSocket('ws://127.0.0.1:5050/ws');
      ws.onopen = () => { this.ws = ws; this.backoff = 1000; };
      ws.onclose = () => { this.ws = null; this.retry(); };
      ws.onerror = () => {}; // onclose follows
      ws.onmessage = (m) => {
        let msg: any; try { msg = JSON.parse(m.data); } catch { return; }
        const done = msg.request_id && this.waiting.get(msg.request_id);
        if (done) { this.waiting.delete(msg.request_id); done(msg); }
      };
    } catch { this.retry(); }
  }

  private retry(): void {
    setTimeout(() => this.connect(), this.backoff);
    this.backoff = Math.min(this.backoff * 2, 30000);
  }

  send(command: object): Promise<Reply> {
    if (!this.ws || this.ws.readyState !== WebSocket.OPEN) {
      return Promise.resolve({ status: 0, code: 'RECORDER_UNAVAILABLE' });
    }
    const request_id = 'cmcd-' + crypto.randomUUID();
    return new Promise((resolve) => {
      this.waiting.set(request_id, resolve);
      setTimeout(() => {
        if (this.waiting.delete(request_id)) resolve({ status: 0, code: 'TIMEOUT' });
      }, 3000);
      this.ws!.send(JSON.stringify({ ...command, request_id }));
    });
  }
}

export const aimscribe = new AimScribe(); // call aimscribe.connect() once, at login

```

B.2 The two calls in the page

```
// When patient details have loaded. visit.start_time came from CMED's server.
aimscribe.send({
  command: 'start',
  trigger: {
    patient_id: visit.patient_id,
    doctor_id:  visit.doctor_id,
    hospital_id: visit.hospital_id,
    start_time: visit.start_time,
    date:       visit.date,
  },
}).then((r) => {
  if (r.code === 'RECORDING_STARTED') visit.session_id = r.data.session_id;
}).catch(() => {});           // SRS-IF1-17: whatever happens, the doctor carries on

// When Build Prescription succeeds.
aimscribe.send({
  command: 'prescription_built',
  patient_id: visit.patient_id,
  session_id: visit.session_id,
  occurred_at: new Date().toISOString(),
}).catch(() => {});
```

B.3 CMED's server: API 2 and the prescription

```
// Runs on CMED's server. The key is read from server configuration only.
const AIMS_URL = process.env.AIMS_URL!;           // given by AIMS LAB
const AIMS_KEY = process.env.AIMS_CMED_KEY!;      // never sent to a browser (SRS-CHB-01)

export function sendToAims(path: string, body: object, attempt = 0): void {
  fetch(`${AIMS_URL}/api/v2/clinical/${path}`, {
    method: 'POST',
    headers: { 'Content-Type': 'application/json', 'X-CMED-Key': AIMS_KEY },
    body: JSON.stringify(body),
    signal: AbortSignal.timeout(10_000),
  })
  .then((r) => {
    if (r.status === 429 || r.status >= 500) throw new Error(String(r.status));
    if (r.status >= 400) console.warn('AIMScribe refused', path, r.status);
  })
  .catch(() => {                                     // SRS-CHB-05: retry for 24 hours
    const delay = Math.min(1000 * 2 ** attempt, 15 * 60_000);
    if (attempt < 110) setTimeout(() => sendToAims(path, body, attempt + 1), delay);
  });
}

// In the handler that returns patient details to the page:
const visit = { patient_id, doctor_id, hospital_id,
  start_time: dhakaTimestamp(),    // created once, here (SRS-CNF-02)
  date: dhakaDate() };
sendToAims('patient-information',          // not awaited (SRS-CHB-06)
  { ...visit, consent_obtained, consent_method, consent_recorded_at,
    demographics, paramedic, previous_visit });
return { ...patientDetails, visit };       // the page uses visit.start_time

// When Build Prescription succeeds:
sendToAims('prescription', { ...visit, issued_at, diagnoses, items,
  investigations, advice, follow_up, notes });
```

A production build should keep unsent requests in a durable queue, so a restart of CMED's server does not lose them.

Appendix C — API reference

C.1 CMED page → recorder (Channel A)

ws://127.0.0.1:5050/ws · JSON · 64 KB maximum · `Origin` required

Command	Sent when	Required fields	Success
start (API 1)	Doctor opens patient details	trigger : patient_id , doctor_id , hospital_id , start_time , date	200 RECORDING_STARTED
prescription_built (API 3)	Prescription built	patient_id , session_id , occurred_at	200 GATE_ARMED
status	Any time	—	Current state
pause · resume · stop	Not used by CMED	—	—

GET `http://127.0.0.1:5050/health` needs no authentication, changes nothing, and is the right way to ask whether the recorder is present.

C.2 CMED server → AIMS LAB server (Channel B)

`https://<aims-lab-server>` · HTTPS · JSON · 1 MB maximum · X-CMED-Key required

Method	Path	Sent when	Success
POST	<code>/api/v2/clinical/patient-information</code> (API 2)	The moment the patient is opened	202 ACCEPTED
POST	<code>/api/v2/clinical/prescription</code> (API 3)	Prescription built	202 ACCEPTED

C.3 Recorder and workers → AIMS LAB server (internal)

Listed so CMED's reviewers can see the whole chain of custody. CMED calls none of these.

Method	Path	Credential	Purpose
POST	/api/v2/device/enroll	Enrolment code	Exchange a one-time code for a device identity
POST	/api/v2/grant/mint	Device token	Permission to record one consultation, and its confirmation (<i>to be built</i>)
POST	/api/v2/session/open	Device token	Open a session, submit chain entry o
POST	/api/v2/segment/authorize	Device token	300-second upload link to R2 for one piece
POST	/api/v2/segment/commit	Device token	Declare uploaded, with SHA-256; server reads it back and checks it
POST	/api/v2/session/pause · /resume	Device token	Pause with its reason
POST	/api/v2/session/close	Device token	Close and submit the end of the chain
GET	/api/v2/session/{id}/receipts	Device token	Collect receipts
POST	/api/v2/heartbeat	Device token	Liveness, buffer depth, version, level figures
GET	/api/v2/doctors	Device token	Doctor names, own clinic only
GET	/api/v2/archive/pending	Worker key	Sessions ready to archive
POST	/api/v2/archive/complete	Worker key	Report an archived session
POST	/api/v2/admin/hospital · /doctor	Admin key	Register a clinic or doctor
POST	/api/v2/admin/enrollment-token	Admin key	Create a single-use enrolment code
POST	/api/v2/admin/device/{id}/revoke	Admin key	Cut off a lost machine
POST	/api/v2/admin/cmcd-key	Admin key	Issue or rotate the CMED API key
GET	/api/v2/admin/alerts	Admin key	Unresolved integrity alerts

Four credentials — `X-Device-Token`, `X-Worker-Key`, `X-Admin-Key` and `X-CMED-Key` — none of which works in place of another (`SRS-IF2-02`).

C.4 Tools for testing

SRS-API-01 [M, D, AIMS] A Postman collection shall cover every route in C.2 and C.3, with an environment file holding addresses and credentials as variables, so no secret is committed with it.

SRS-API-02 [M, D, AIMS] The collection shall include a working end-to-end sequence — enrol, grant, open, upload link, commit, close, receipts — that a reviewer can run against the test environment without reading code.

SRS-API-03 [M, I, AIMS] The AIMS LAB server shall publish OpenAPI at `/openapi.json` and interactive documentation at `/docs`.

SRS-API-04 [S, D, AIMS] A test page shall be provided to CMED: one HTML file that connects to `127.0.0.1:5050`, sends `start` and `prescription_built`, and prints the replies. Postman cannot test the loopback connection, so this is the first tool CMED's developer will use.

SRS-API-05 [M, D, AIMS] A test Channel B environment shall be provided, with its own address and key, and a page showing CMED's developer each record received, whether it passed validation, and whether it confirmed a recording.

Appendix D — Requirement index

Prefix	Area	Section
INV	The three identities	2.2
USR	Users	2.3
TOP	Connections and trust	3.2, 3.3, 3.6
ENR	Enrolment	4
GRT	Grants	5.1–5.5
CNF	Confirmation with CMED's server	5.6
IF1	Channel A — CMED page to recorder	6.1
CHB	Channel B — CMED server to AIMS LAB server	6.2
IF2	Recorder to AIMS LAB server	6.3
STO	Cloud storage	6.4
IF3	Workers	6.5
CAP · LVL	Capture and speech levels	7.1, 7.1a, 7.1b
SEG · SPL · CHN · UPL	Pieces, buffer, chain, upload	7.2–7.5
REC	Failed checks and deletion	7.5a
SES · GAT · UIX	Sessions, gate, on-screen control	7.6–7.8
BKD · ARC	Server, archive and cloud copy	7.9, 7.10
DAT	Data	8.1–8.5
CRI	Receiving clinical records	8.6
DBA	The two databases	8.7
DSH	Dashboard	8.9
NFP · NFC · NFR · NFS · NFO · NFM · NFU	Non-functional	9
SRV	The UIU server	10.3
API	Testing tools	Appendix C.4

Appendix E — Document history

Version	Date	Change
3.1	14 September 2026	Consent travels in API 2 (<code>consent_obtained</code> , <code>consent_method</code> , <code>consent_recorded_at</code>), closing OD-13. Without consent the recording stops, captured audio is deleted without upload, and no clinical content is kept (<code>SRS-CNF-11 – 12</code> , <code>SRS-UIX-14</code> , reply <code>200 CONSENT_NOT_GIVEN</code>). <code>SRS-BKD-01</code> , <code>SRS-GRT-03</code> and <code>SRS-GRT-08</code> updated; failure 23 and <code>AT-77</code> added; <code>AT-03</code> rewritten.
3.0	14 September 2026	Whole document brought in line with the plan of 13 September 2026, and rewritten for CMED readers. AIMS LAB server hosted at UIU (§10.3, <code>SRS-SRV-01 – 09</code> ; OD-10 closed). CMED's three signals over two channels described end to end, with Channel B defined as an interface (§6.2, <code>SRS-CHB-01 – 12</code>). Trigger reduced to five fields; arm signal renamed <code>prescription_built</code> . Audio journey from PC to archive and cloud copy (§3.5); cloud storage requirements (<code>SRS-STO-01 – 06</code>) and sizing (§10.4). Two databases and the JSON file beside each recording (§8.7, §8.8). <code>SRS-ARC-05</code> withdrawn; <code>SRS-ARC-13 – 16</code> , <code>SRS-UIX-13</code> , <code>SRS-DSH-08 – 09</code> , <code>SRS-NFR-06</code> , <code>SRS-API-05</code> added. Failure modes extended to 22; tests <code>AT-68 – AT-76</code> added. New figures 1 and 6; figures that described the earlier plan replaced. Open decisions OD-13 to OD-21 added.
2.3	13 September 2026	Cloud pieces no longer simply deleted after archiving: the recording is stored and verified at UIU first, a merged lossless FLAC copy is verified and kept in a locked bucket, and only then are the pieces deleted (<code>SRS-ARC-08 – 12</code> , <code>SRS-DAT-17</code> , <code>AT-64 – AT-67</code>).
2.2	12 September 2026	Trigger confirmation (<code>SRS-CNF-01 – 10</code>): the trigger stays on the PC, and each recording is confirmed against the patient information CMED's server sends at the same moment. Tests <code>AT-57 – AT-63</code> .
2.1	11 September 2026	Why audio stayed on PCs: receipts waited for archiving, then a 24-hour wait expired while PCs were off. <code>SRS-REC-15 – 18</code> issue receipts on custody, remove the wait, and clean up at start-up and shutdown.
2.0	10 September 2026	Plain-language reading guide. Channel B added: CMED sends clinical data directly to the AIMS LAB server. Rule that nothing is stored on the doctor's PC.
1.8	10 September 2026	Buffer sized for minutes (4 GB), not weeks. Recording catalogue and clinical record split into two databases with separate credentials.
1.7	9 September 2026	Quarantined material delivered like any other; nothing waits on a PC for a person (<code>SRS-REC-10 – 14</code>). Search problems re-diagnosed (<code>SRS-DBA-22 – 25</code>).
1.6	9 September 2026	Clinical record ingestion, database architecture and the dashboard added; consent and access requirements <code>SRS-DAT-15 – 16</code> .
1.5	5 September 2026	Prepared for submission; diagrams redrawn as figures.
1.4	25 August 2026	The recorder already re-checks every piece before upload; all five quarantine causes listed.
1.3	25 August 2026	Recovery of pieces that fail their check (<code>SRS-REC-01 – 09</code> , <code>AT-42 – AT-46</code>).

Version	Date	Change
1.2	25 August 2026	Figures checked for label collisions; API reference and testing tools; CMED shows nothing, all doctor messages come from the on-screen control.
1.1	25 August 2026	Sizing corrected to 14 rooms; capture path, speech levels, original and derived copies, clinic register.
1.0	25 August 2026	First baseline for the CMED integration meeting.

Questions, corrections and disagreements are welcome. If anything in this document is unclear, that is a fault in the document — tell AIMS LAB and it will be fixed, rather than leaving CMED to guess.

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